Title of the Thesis

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Position, Department,
Institution

Month, Year

To my family.

Acknowledgments

I would like to thank my supervisor, **Prof. Supervisor Name**, for the guidance and support throughout this work.

Resumo

Este trabalho aborda o problema de …

Palavras-chave: palavra-chave 1, palavra-chave 2, palavra-chave 3.

Abstract

This thesis addresses the problem of ...

Keywords: keyword 1, keyword 2, keyword 3.

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CHAPTER 1

Introduction

This chapter introduces the addressed problem, motivates the work, states the research questions, summarises the contributions, and outlines the remainder of the document.

1.1. Motivation

1.2. Problem Statement

1.3. Research Questions

This work addresses the following research questions:

RQ1.

RQ2.

RQ3.

1.4. Objectives

The main objectives are:

-
-

1.5. Contributions

The main contributions are:

-
-

1.6. Document Structure

The remainder of this document is organised as follows. Chapter 2 reviews the relevant literature and prior work. Chapter 3 describes the proposed approach. Chapter 4 presents and discusses the experimental results. Chapter 5 summarises the work and outlines directions for future work.

CHAPTER 2

Literature Review

This chapter presents the review methodology used to identify relevant literature and synthesises the related work on three-dimensional left-ventricle reconstruction from cardiac MRI and myocardial wall-thickness measurement.

2.1. Review Methodology

The literature search followed a structured approach to ensure broad coverage and relevance. This section describes the search strategy, selection criteria, and results.

2.1.1. Search strategy

The search was conducted across three major bibliographic databases: Scopus, PubMed, and IEEE Xplore, supplemented by Google Scholar for citation chaining. The temporal scope covered publications from 2000 to 2026, prioritising work from the last decade while retaining foundational methods essential for theoretical context.

Search terms were grouped into four thematic categories and combined using Boolean operators. The first category covered anatomy and imaging terms (“cardiac MRI”, “left ventricle”, “short-axis”, “SAX”, “myocardium”). The second addressed reconstruction (“3D reconstruction”, “mesh reconstruction”, “shape model”, “implicit surface”, “signed distance function”). The third targeted learning methods (“deep learning”, “graph neural network”, “point cloud”, “neural representation”). The fourth focused on measurement (“wall thickness”, “myocardial thickness”, “Laplace equation”, “transmural”).

The refined query applied consistently across databases was:
(“cardiac MRI” OR “left ventricle” OR “short-axis”) AND (“3D reconstruction” OR “statistical shape model” OR “signed distance function” OR “graph neural network” OR “implicit representation”) AND (“wall thickness” OR “mesh” OR “deep learning”)

2.1.2. Inclusion and exclusion criteria

Studies were included if they: (i) addressed three-dimensional reconstruction or shape modelling of the left ventricle, (ii) proposed or evaluated methods for myocardial wall-thickness

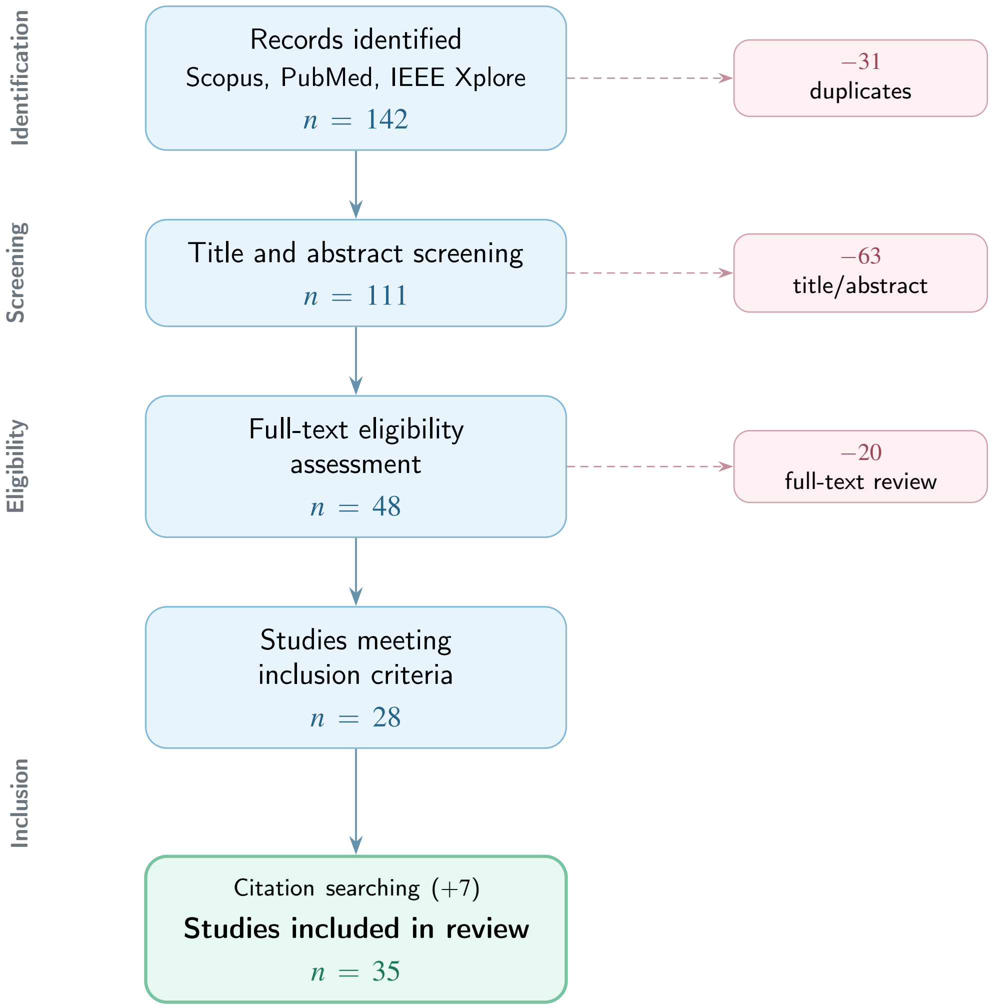

Figure 2.1. PRISMA flow diagram summarising the literature selection process.

measurement, (iii) developed geometric deep learning techniques applicable to anatomical meshes, or (iv) introduced benchmark cardiac MRI datasets used for LV analysis.

Studies were excluded if they: (i) focused exclusively on segmentation without reconstruction or shape analysis, (ii) addressed organs other than the heart without methodological generalisability, (iii) lacked peer review or sufficient methodological description, or (iv) were published before 2000 unless they constituted foundational work directly relevant to the methods reviewed.

2.1.3. Results

The initial search retrieved 142 records across the three databases. After removing 31 duplicates, 111 unique records were screened by title and abstract. Of these, 48 were selected for full-text assessment. Following detailed review against the inclusion criteria, 28 studies were retained. An additional 7 studies were identified through backward citation searching, yielding a final corpus of 35 publications.

The selected studies span five thematic areas: cardiac MRI acquisition and segmentation, statistical shape models, implicit neural representations, graph neural networks for mesh processing, and wall-thickness measurement. The following section synthesises the key findings from this corpus.

2.2. Related Work

This section synthesises the reviewed literature, organised by thematic area.

2.2.1. Cardiac MRI segmentation and datasets

Short-axis (SAX) cine MRI is the standard protocol for left-ventricle (LV) evaluation, producing 2D slices perpendicular to the long axis with high in-plane resolution (1–2 mm) but large inter-slice gaps (6–10 mm) (Cerqueira et al., 2002; Petitjean & Dacher, 2011). This anisotropic acquisition creates the fundamental challenge that motivates 3D reconstruction: the ventricle must be inferred volumetrically from sparse planar observations. Petitjean and Dacher (2011) reviewed over 100 segmentation methods and concluded that the inter-slice gap remains the single largest source of error in volumetric cardiac analysis, with apex and base delineation accounting for the majority of inter-observer variability.

The American Heart Association (AHA) 17-segment model, introduced by Cerqueira et al. (2002), standardised the partition of the myocardium into basal, mid-cavity, and apical regions. This model is universally adopted for reporting wall motion, perfusion, and thickness, and serves as the reference frame for all regional analysis discussed in subsequent sections.

Segmentation of endocardial and epicardial contours is a prerequisite for any downstream 3D analysis. Ronneberger et al. (2015) proposed U-Net, an encoder–decoder architecture with skip connections that was originally designed for neuronal structure segmentation but rapidly became the de facto backbone for cardiac MRI analysis. The architecture trains effectively from small annotated datasets and produces pixel-level predictions that, when applied to SAX slices, yield endocardial and epicardial contours with sub-pixel accuracy.

The Automated Cardiac Diagnosis Challenge (ACDC), organised by Bernard et al. (2018), provided the first standardised cardiac segmentation benchmark. The dataset comprises 100 patients distributed across five clinical groups: normal, dilated cardiomyopathy (DCM), hypertrophic cardiomyopathy (HCM), abnormal right ventricle, and previous myocardial infarction. Each case includes expert delineations at end-diastole (ED) and end-systole (ES). The challenge reported that top-performing deep learning methods achieved mean Dice scores exceeding 0.93

for the LV cavity and 0.89 for the myocardium, demonstrating that automated segmentation had reached expert-level performance for standard acquisitions.

Isensee et al. (2021) introduced nnU-Net, a self-configuring framework that automatically adapts network topology, patch size, preprocessing, and post-processing to each dataset. Applied to the ACDC benchmark, nnU-Net matched or exceeded all prior challenge submissions without any manual architecture design. The framework has since become the standard baseline against which new cardiac segmentation methods are evaluated.

The Multi-Centre, Multi-Vendor and Multi-Disease (M&Ms) challenge, described by Campello et al. (2021), extended evaluation to the clinically critical question of domain generalisation. The dataset includes 345 subjects acquired on scanners from four different vendors (Siemens, General Electric, Philips, and Canon) across six clinical centres. The challenge demonstrated that while methods trained on a single vendor achieved high accuracy on that vendor, performance dropped significantly when tested on unseen vendors — with Dice reductions of 5–15 percentage points in some cases. This finding motivated the development of domain-adaptation techniques and highlighted the importance of multi-source training data.

For population-level analysis, the UK Biobank (Sudlow et al., 2015) provides the largest available resource, with cardiac MRI acquired for over 40 000 participants using a standardised 1.5T protocol. Bai et al. (2018) developed a fully convolutional pipeline to segment this cohort at scale, producing automated delineations that, after quality control, serve as the foundation for large-scale shape modelling and epidemiological studies.

2.2.2. Statistical shape models

Statistical Shape Models (SSMs) encode anatomical variability by aligning a training population to point-to-point correspondence and applying PCA to extract the dominant modes of variation (Cootes et al., 1995; Heimann & Meinzer, 2009). Cootes et al. (1995) introduced Point Distribution Models, where a training set of shapes — each represented by k corresponding landmarks — is rigidly aligned via Procrustes analysis, averaged into a mean shape $\bar{x}$, and decomposed via PCA into principal modes Φ. A new shape is generated as $x = \bar{x} + \Phi b$, where b are shape parameters constrained within $\pm 3\sigma$ of each eigenvalue to remain anatomically plausible. In their experiments on hand and face contours, Cootes et al. (1995) showed that the first 5–10 modes captured over 95% of population variance, demonstrating the efficiency of the linear representation.

Heimann and Meinzer (2009) surveyed over 100 studies applying SSMs to medical image segmentation and identified the cardiac domain as particularly well-suited because of the regularity of ventricular geometry and the availability of large training cohorts. Their review highlighted that SSM accuracy depends critically on three factors: the quality of inter-subject correspondence, the size and diversity of the training set, and the alignment method used.

Frangi et al. (2002) pioneered automatic construction of multi-object cardiac SSMs, proposing a non-rigid registration framework that establishes dense correspondence across subjects without manual landmarks. Applied to biventricular models, their approach captured coupled shape variation between the left and right ventricles, demonstrating that multi-structure models outperform single-structure ones for segmentation-by-fitting tasks.

Bai et al. (2015) constructed the most comprehensive cardiac atlas to date, using over 1000 high-resolution MRI scans from the UK Digital Heart Project. Their biventricular model captures both end-diastolic shape and wall motion through the cardiac cycle. The authors reported that the first 50 PCA modes explain approximately 90% of shape variance, and demonstrated that the atlas enables automatic segmentation of new subjects by fitting the model to image evidence with mean surface-to-surface errors below 1.5 mm.

de Marvao et al. (2014) applied the same UK Biobank-derived SSM to study myocardial hypertrophy and showed that atlas-based shape analysis detects subtle remodelling patterns that volumetric indices alone (such as LV mass) miss. Their study demonstrated substantially improved statistical power for detecting hypertrophic phenotypes when using shape modes compared to traditional clinical measurements, requiring fewer subjects to achieve equivalent sensitivity.

Medrano-Gracia et al. (2014) used a multi-ethnic cohort of over 1900 subjects to characterise normal LV shape variation. They reported that the dominant modes correspond to overall ventricular size, followed by eccentricity, apical shape, and regional wall curvature — confirming that cardiac shape cannot be adequately described by a single size parameter. Suinesiaputra et al. (2018) organised an international challenge to classify myocardial infarction from SSM-derived shape features, showing that machine learning classifiers trained on the first 20 shape modes achieved AUC values above 0.85 for distinguishing infarcted from healthy ventricles.

SSMs serve dual roles in reconstruction pipelines. First, they act as *geometric priors* that regularise shape recovery from sparse observations, constraining the solution to lie on or near the learned manifold. Second, they function as *synthetic data generators*: by sampling the latent space, one can produce thousands of anatomically plausible meshes for training data-hungry deep learning models without additional manual annotation (Heimann & Meinzer, 2009; Khalafvand et al., 2018). Khalafvand et al. (2018) demonstrated this second role by generating

SSM-sampled LV geometries to train computational fluid dynamics simulations, showing that the synthetic population reproduced the haemodynamic statistics of real subjects.

2.2.3. Deep learning models for 3D shape reconstruction

Classical reconstruction methods interpolate between segmented SAX slices using splines or deformable surfaces, but cannot recover geometric detail between slices, particularly near the apex and base (Heimann & Meinzer, 2009; Petitjean & Dacher, 2011). Petitjean and Dacher (2011) noted that linear interpolation between contours produces visible staircase artefacts and fails to reproduce the smooth taper of the LV apex, while parametric models (prolate spheroids, B-spline surfaces) impose smoothness but cannot express patient-specific morphological detail. Modern approaches replace interpolation with learned shape recovery, falling into two broad families: template deformation and implicit field prediction.

Template deformation methods maintain a fixed mesh topology and learn per-vertex displacements conditioned on input observations. Wang et al. (2018) proposed Pixel2Mesh, which takes a single RGB image as input and iteratively refines an ellipsoidal template mesh through three stages of graph convolutional layers. At each stage, vertex features are enriched with image features sampled via learned projections, and the mesh is refined through unpooling. The authors reported Chamfer distances below 0.5 on the ShapeNet benchmark, demonstrating that graph-based deformation can recover complex topology from minimal input. Choi et al. (2020) extended this paradigm with Pose2Mesh, which recovers full 3D human body meshes from sparse 2D pose keypoints. Their cascaded coarse-to-fine architecture first predicts a coarse mesh from joint positions and then refines vertex locations with finer graph convolutions, achieving state-of-the-art results on the Human3.6M benchmark. The architectural principle — recovering dense 3D geometry from sparse 2D observations — is directly analogous to the cardiac problem of inferring a full LV mesh from a handful of 2D SAX contours.

In the cardiac domain, template deformation preserves the topological consistency required for cross-subject comparison and downstream analysis such as wall-thickness computation (Bai et al., 2015; Suinesiaputra et al., 2014). Beetz et al. (2021) applied point-cloud variational autoencoders to biventricular shapes from the UK Biobank, learning a compact latent space that enables generation of subpopulation-specific anatomy models. Their decoder produces meshes with fixed vertex correspondence, making population statistics directly computable from the generated samples.

Implicit Neural Representations (INRs) take a different approach. Rather than deforming a template, they encode geometry as a continuous function whose zero-level set defines the surface. Park et al. (2019) introduced DeepSDF, which parameterises the signed distance function (SDF) of an entire shape class with a single neural network conditioned on a per-shape latent code. Given partial or noisy point-cloud observations, their auto-decoder architecture optimises the latent code at test time to find the shape in the learned space that best explains the input. On the ShapeNet benchmark, DeepSDF achieved reconstruction quality comparable to voxel-based methods while using an order of magnitude fewer parameters, and naturally supports shape interpolation in latent space. Mescheder et al. (2019) proposed Occupancy Networks, which predict binary inside/outside occupancy rather than signed distance. Their encoder processes a point cloud or image to produce a latent vector, and the decoder evaluates occupancy at arbitrary 3D query points. Both DeepSDF and Occupancy Networks produce resolution-independent surfaces extracted via the Marching Cubes algorithm (Lorensen & Cline, 1987), which generates a watertight triangulated mesh from the zero-crossing of the implicit field on a regular grid.

Two techniques are critical for applying INRs to thin cardiac structures where high-frequency geometric detail must be preserved. Tancik et al. (2020) demonstrated that standard MLPs exhibit spectral bias — they preferentially learn low-frequency components — and proposed mapping input coordinates through random Fourier features before the network. In their experiments on 2D image regression and 3D shape reconstruction, Fourier encoding substantially reduced reconstruction error compared to raw coordinate input, with the improvement most pronounced for high-frequency details such as thin structures and sharp edges. Gropp et al. (2020) introduced Implicit Geometric Regularisation, showing that enforcing the eikonal constraint ($\|\nabla f\| = 1$) as a loss term during training produces valid distance fields without requiring ground-truth SDF values at every point. Their approach enables training from raw point clouds alone, which is particularly relevant when dense SDF supervision is expensive to compute.

For encoding unordered input observations such as SAX contour points, the PointNet architecture (Qi et al., 2017) provides a permutation-invariant encoder via shared MLPs followed by symmetric max-pooling. Qi et al. (2017) demonstrated that this simple design achieves classification accuracy of 89.2% on the ModelNet40 benchmark and produces compact global descriptors that capture the essential geometry of the input point set regardless of its size or ordering. In cardiac applications, each SAX slice contributes a set of contour points that can be processed by a PointNet-style encoder to produce a latent representation of the observed geometry.

2.2.4. Wall-thickness measurement techniques

Myocardial wall thickness is a primary clinical biomarker for cardiac remodelling. Grossman et al. (1974) demonstrated that wall thickness is a principal determinant of left-ventricular diastolic stiffness, with hypertrophied ventricles exhibiting markedly elevated stiffness proportional to end-diastolic wall thickness. More recently, Huelnhagen et al. (2017) showed at 7.0T MRI that wall thickness correlates with tissue transverse relaxation properties, confirming its value as a non-invasive surrogate for myocardial tissue characterisation. The AHA 17-segment model (Cerqueira et al., 2002) provides the standard framework for reporting regional thickness values.

The accuracy of wall-thickness measurement depends critically on the quality of the underlying surface reconstruction, as any smoothing or geometric artefact propagates directly into the thickness estimate. The literature describes five families of measurement methods, each with distinct trade-offs.

The simplest approach uses nearest-neighbour distance: for each endocardial vertex, a KD-tree identifies the closest epicardial point, yielding a measurement in $O(N \log M)$ time. While computationally trivial (under 1 ms for typical cardiac meshes), this method provides only a lower bound on the true transmural thickness because it ignores the anatomical direction of measurement — the nearest point may lie tangentially rather than across the wall.

Ray-based methods address this limitation by casting a ray from each endocardial vertex along its surface normal until it intersects the epicardium. The intersection is computed via the Möller–Trumbore algorithm, accelerated with a Bounding Volume Hierarchy (BVH) to $O(\log F)$ per query. This approach measures transmurally but is sensitive to normal noise: a slightly rotated normal can produce a spurious long or short ray. The Shape Diameter Function variant improves robustness by casting a cone of K rays (typically $K{=}7$) at a half-angle of 30° around the normal and reporting the median intersection distance, effectively filtering outlier hits caused by local mesh irregularities.

Volumetric methods operate on voxelised representations. The Euclidean Distance Transform (EDT) computes, for each myocardial voxel, the distance to the nearest boundary, and the sum of the endocardial and epicardial distance fields approximates local thickness ($t = D_{\text{endo}} + D_{\text{epi}}$). This estimate is exact for planar parallel boundaries and overestimates by approximately 5–10% for the curvatures typical of the LV mid-wall. The method is computationally efficient ($O(V)$ via the Saito–Toriwaki algorithm) but limited by voxel resolution.

PDE-based methods represent the gold standard for transmural thickness. Jones et al. (2000) introduced the Laplacian approach — originally for cortical thickness in brain imaging — which

[Placeholder] Summary diagram of wall-thickness measurement families: (a) KD-tree nearest neighbour, (b) normal ray, (c) Laplace streamlines, (d) AHA-17 bullseye output.

Figure 2.2. Overview of wall-thickness measurement approaches, ranging from simple distance queries to PDE-based transmural streamlines, with results aggregated into the standardised AHA 17-segment bullseye.

solves $\nabla^2\psi = 0$ with Dirichlet conditions (ψ=0 at the endocardium, ψ=1 at the epicardium). The uniqueness of the harmonic solution guarantees that the resulting streamlines never cross, producing a globally consistent transmural direction field. The local thickness is $t = 1/|\nabla\psi|$. While conceptually elegant, this formulation amplifies numerical errors where the gradient magnitude is small (thin walls, apex). Yezzi and Prince (2003) proposed an improvement: instead of dividing by $|\nabla\psi|$, they solve two advection equations ($\nabla\psi \cdot \nabla u = -1$ from the endocardium and $\nabla\psi \cdot \nabla v = -1$ from the epicardium), with the total thickness given directly as $t = u + v$. This Eulerian PDE formulation avoids the numerical instability of gradient division and produces identical results to the Laplacian method in smooth regions while being significantly more stable near the apex and in walls thinner than 4 mm.

Finally, mesh-based regularised correspondence methods combine an initial geometric estimate (normal projection or nearest neighbour) with iterative Laplacian smoothing on the mesh connectivity graph. At each iteration, each vertex's thickness is updated as a weighted average of its neighbours, anchored to the initial estimate. This produces spatially smooth thickness maps suitable for bullseye visualisation, at the cost of potentially attenuating real local thickness variation if the smoothing parameter is set too aggressively.

2.2.5. Summary and research gap

The reviewed literature shows that each component of the LV analysis pipeline is well-studied in isolation: segmentation is effectively solved by deep learning, SSMs encode population-level shape priors, implicit representations provide continuous surface modelling, and multiple wall-thickness algorithms exist. However, few studies integrate these components into a single pipeline for the specific setting addressed in this thesis. The missing combination is a method that encodes sparse SAX contour points, decodes paired endocardial and epicardial surfaces as implicit signed distance fields, conditions the reconstruction on cardiac phase, guarantees positive wall thickness by construction, and trains jointly on real clinical data and SSM-derived synthetic samples. This gap motivates the approach described in the remaining chapters.

CHAPTER 3

Methodology

This chapter describes the methodology used to reconstruct 3D LV geometry from sparse short-axis contours and to measure myocardial wall thickness on the reconstructed model. The proposed pipeline combines synthetic and real cardiac MRI-derived data, a phase-conditioned implicit neural representation, and a systematic comparison of local wall-thickness algorithms.

3.1. Overview

The complete thesis workflow can be read as a progression from sparse clinical observations to continuous three-dimensional geometry and then to quantitative wall-thickness summaries. It starts from segmented short-axis (SAX) cardiac MRI stacks or synthetic Statistical Shape Model (SSM) meshes. In both cases, the available geometry is converted into a common contour-based representation composed of endocardial and epicardial points sampled across SAX slices. These points are passed to a PointNet-style encoder that produces a compact latent code for the observed LV geometry. A signed-distance implicit decoder then predicts two coupled surfaces: the endocardium and the epicardium. The epicardial field is parameterised as a monotone offset from the endocardial field, which guarantees positive wall thickness by construction. The resulting surfaces are finally used to compute local thickness maps and regional AHA-17 summaries.

The methodological design is organised around four objectives. First, the input representation must be compatible with both synthetic and real cases. Second, the model must reconstruct smooth watertight LV surfaces from sparse 2D observations. Third, the decoder must produce a valid myocardial wall everywhere, not only at the observed slices. Fourth, the reconstructed geometry must support local wall-thickness measurements that can be visualised as surface colour maps and regional AHA-17 bullseye plots.

The chapter is therefore organised as a technical pipeline rather than as a single model description. Section 3.2 explains how synthetic and real data were converted into a shared cache representation. Section 3.3 describes the model encoder and decoder, with particular attention to the monotone-epi parameterisation used to guarantee positive wall thickness. Section 3.4 presents the training and inference procedure, and Section 3.5 describes the wall-thickness algorithms used to evaluate local measurements on the reconstructed geometry.

Figure 3.1 summarises this end-to-end workflow and should be interpreted as an overview of the thesis methodology rather than as a figure belonging only to data preparation. The first panels illustrate the sparse SAX observation and contour extraction, the mesh panel represents the reconstructed LV surface domain, and the lower panels show how the reconstructed geometry is converted into local and regional wall-thickness measurements. In this preview, panels (a) and (b) are drawn from the ACDC Challenge dataset, whereas panels (c), (d), and (e) are drawn from the UK Biobank LV SSM pipeline.

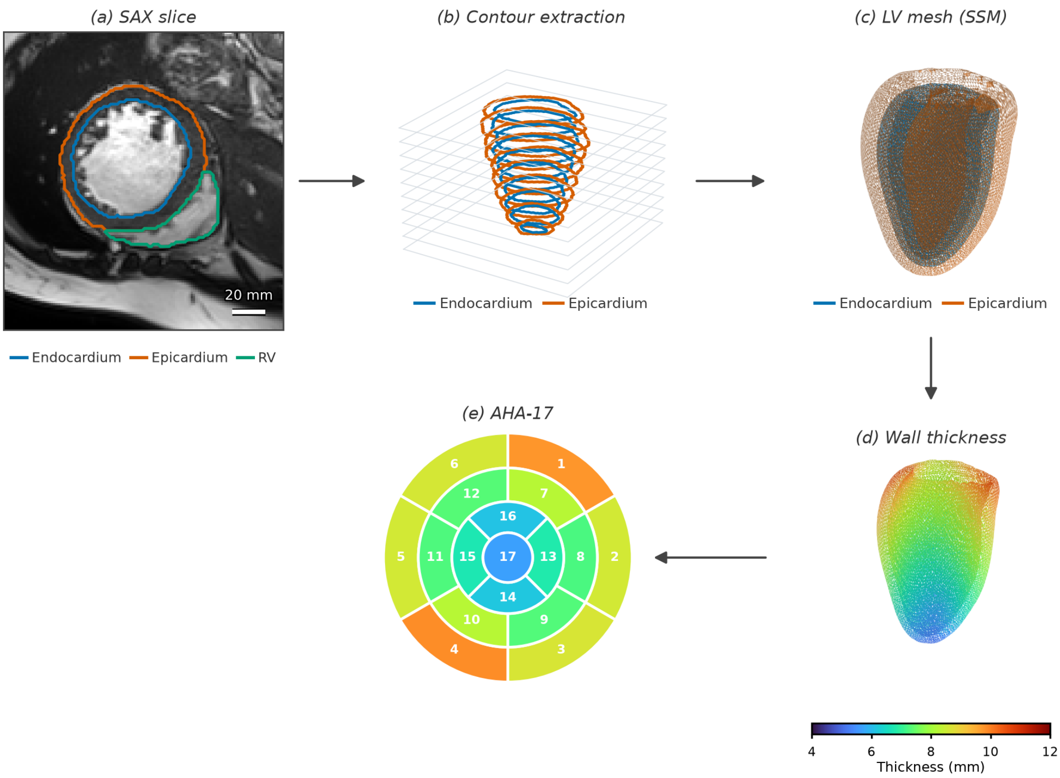

Figure 3.1. Preview of the end-to-end methodology from sparse SAX observations to reconstructed LV geometry and wall-thickness outputs.

The chapter then follows this same progression in prose. Data construction and cache standardisation are detailed in Section 3.2. The model representation is developed in Section 3.3, where the contour encoder and monotone-epi implicit decoder are formalised. The optimisation and inference pipeline are presented in Section 3.4, and the thickness-analysis protocol, including local maps and AHA-17 aggregation, is described in Section 3.5.

3.2. Data Preparation

A fundamental challenge in training the model is the absence of a paired dataset that provides both sparse SAX contour observations *and* corresponding validated three-dimensional

LV meshes as ground truth. Clinical cardiac MRI datasets supply segmentation masks and image volumes, but they do not include the watertight endocardial and epicardial surfaces that an SDF decoder requires as supervision targets. This missing ground-truth problem is the starting point for the entire data-preparation strategy.

To address this issue, a two-stage approach was used. In the first stage, 1300 synthetic ED meshes were generated from the UK Digital Heart Project Statistical Shape Model (SSM) of the LV, providing anatomically plausible three-dimensional geometry with controlled shape variation. The model was pre-trained on this synthetic corpus, which established a geometric prior for LV shape variability. In the second stage, the training corpus was extended with meshes reconstructed directly from real clinical segmentations: ED and ES frames from the ACDC, M&Ms, and M&Ms-2 datasets. This fine-tuning stage introduced patient-specific anatomy, acquisition variability, and phase diversity that the SSM alone cannot provide.

All three data streams were converted into the same cache schema. The shared format allows the pre-training and fine-tuning stages to use the same training loop and loss function without any dataset-specific logic.

3.2.1. The missing ground-truth problem and motivation for SSM

The limitation outlined above can be stated more precisely as a supervision-gap problem: the model requires paired samples of sparse SAX contours and reliable three-dimensional LV meshes, while public cine MRI benchmarks provide mainly slice-wise labels. In datasets such as ACDC (Bernard et al., 2018), M&Ms (Campello et al., 2021), and M&Ms-2 (Martín-Isla et al., 2023), the available annotations are segmentation masks rather than watertight endocardial and epicardial surfaces suitable for direct SDF supervision.

Two complementary strategies were therefore adopted to create the missing supervision. First, a Statistical Shape Model was used to generate a large number of synthetic ED meshes that are anatomically valid by construction. This solved the ground-truth problem for the pre-training stage at the cost of not capturing real segmentation artefacts or scanner-specific anatomy. Second, real ED and ES segmentation masks were converted into watertight three-dimensional meshes through a controlled preprocessing pipeline. This provided the patient-specific geometry needed for fine-tuning, at the cost of requiring careful mesh reconstruction from inherently discrete voxel data.

The SSM pre-training approach is motivated by the observation that learning the geometry of the LV, rather than fitting individual segmentations, requires exposure to the full range of normal and pathological LV shapes. The SSM encodes this range compactly as a mean shape and a set of principal modes of variation, and sampling from its latent space produces diverse but anatomically constrained meshes. The real-data fine-tuning stage then ensures that the model generalises to the segmentation quality and acquisition conditions present in the evaluation datasets.

3.2.2. Statistical Shape Model of the left ventricle

A Statistical Shape Model represents the variability of an anatomical structure across a population by applying principal component analysis (PCA) to a set of aligned training meshes (Heimann & Meinzer, 2009). Each mesh is expressed as a vector of vertex coordinates; the mean of all training shapes forms the mean shape $\bar{X}$, and the eigenvectors of the covariance matrix of the centred shapes form the principal modes. Any shape in the population can be approximated as a linear combination of these modes, and new shapes that follow the observed statistics can be generated by sampling the latent coefficient vector.

This thesis uses the UK Digital Heart Project LV SSM (Bai et al., 2015), derived from UK Biobank cine MRI data of healthy volunteers. The model provides a mean endocardial surface of 22 043 vertices together with 100 principal modes of variation stored as a 66 129 $\times$ 100 matrix of mode vectors and a corresponding vector of 100 eigenvalues λ_i. The standard deviations $\sigma_i = \sqrt{\lambda_i}$ quantify the magnitude of each mode. A new endocardial mesh is generated by

$$X(b) = \bar{X} + \Phi b, \tag{3.1}$$

where $\bar{X}$ is the mean shape flattened to a vector, Φ is the matrix of mode vectors, and $b = [b_1, \ldots, b_{100}]^\top$ is the sampled coefficient vector.

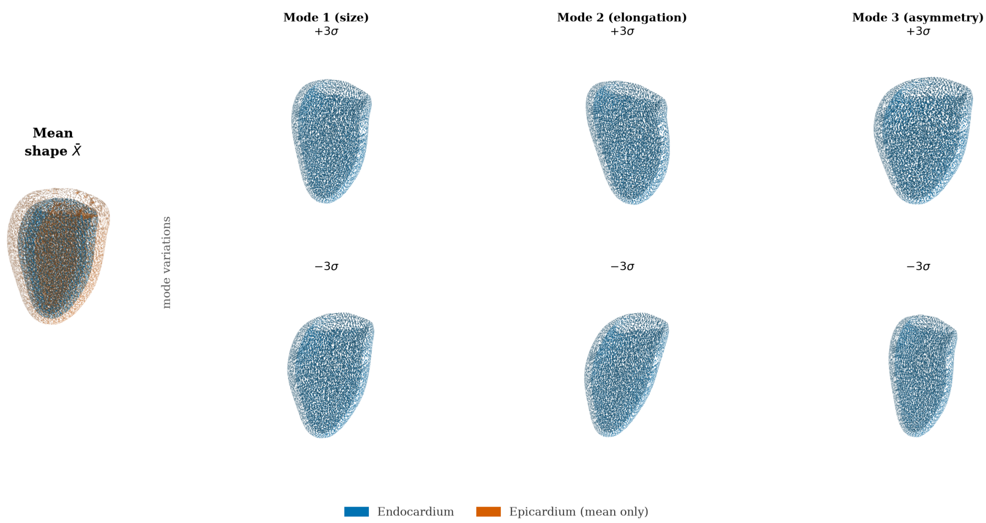

Figure 3.2. UK Digital Heart Project LV SSM mean shape and first three PCA modes at $\pm 3\sigma$.

The first few modes capture the largest sources of shape variation in the population; mode 1 primarily controls overall LV size, mode 2 encodes elongation along the long axis, and higher modes represent progressively more localised deformations, as shown in Figure 3.2.

3.2.3. Synthetic end-diastolic dataset (SSM pre-training)

The synthetic ED dataset was built by sampling 1300 meshes from the SSM and converting each mesh into a training-compatible sample. The generation pipeline proceeds in five steps, illustrated in Figure 3.3.

Step 1: Latent coefficient sampling. Each candidate coefficient vector b was drawn by sampling each component as $b_i \sim \mathcal{N}(0, \sigma_i^2)$ and then applying two acceptance criteria:

$$\left|\frac{b_i}{\sigma_i}\right| \leq 3, \qquad \sum_i \left(\frac{b_i}{\sigma_i}\right)^2 \leq \chi^2_{0.99,d}, \tag{3.2}$$

where σ_i is the standard deviation of mode i, d is the number of active modes, and $\chi^2_{0.99,d}$ is the 99th percentile of the chi-squared distribution with d degrees of freedom. The per-mode clipping prevents individual modes from dominating the shape, and the Mahalanobis bound prevents the combined deformation from falling outside the plausible population distribution. Accepted samples are scattered within a bounded ellipsoid in latent space, as shown in panel (c) of Figure 3.3.

Step 2: Mesh generation. Each accepted coefficient vector b is substituted into Equation (3.1) to produce a deformed endocardial vertex array, which is then triangulated using the face connectivity of the mean-shape mesh. The resulting triangular surface is watertight and oriented by construction.

Step 3: Quality filtering. Before a generated mesh is accepted into the training set, it is evaluated against four proxy metrics: enclosed volume (mL), surface area (cm^2), sphericity index, and plausibility of the apical and basal normal directions. Samples that fall outside calibrated bounds for any of these metrics are discarded. Periodic exact audits using VTK mass-properties computation were also performed every 40 samples to verify the proxy metric calibration.

Step 4: Synthetic epicardium. The SSM provides only the endocardial surface. The epicardial surface was synthesised by offsetting each endocardial vertex p_i along its outward normal n_i by a spatially varying distance d_i,

$$q_i = p_i + d_i n_i, \tag{3.3}$$

where d_i was set to approximately 10 mm near the base and 5 mm near the apex, with additive Gaussian noise of standard deviation 0.5 mm to avoid perfectly uniform geometry. This synthetic epicardium is not intended to reproduce clinical wall thickness but provides a consistent outer surface for SDF supervision during pre-training.

Step 5: SAX contour extraction and volumetric supervision points. Each pair of endocardial and epicardial surfaces was sliced at 10 evenly spaced axial levels to produce sparse SAX contour rings, matching the acquisition pattern of real cine MRI. For each case, 2048 query points were sampled near the surfaces and in the surrounding volume, and signed-distance targets or inside/outside labels were computed. These query points form the dense volumetric supervision used by the SDF loss terms during training.

Figure 3.3 summarises this process in two rows. The top row provides the statistical view of sampling: panel (a) shows the Mahalanobis-statistic density and the $\chi^2_{0.99,32}$ acceptance threshold, panel (b) shows that 32 active modes capture 99% cumulative shape variance, panel (c) shows accepted latent samples in the first two dimensions, and panel (d) shows the synthetic epicardial-offset distribution from Equation (3.3). The bottom row provides the geometric view: panel (e) shows mesh synthesis via Equation (3.1), panel (f) contrasts an accepted shape with a rejected outlier, panel (g) shows the 10 SAX contour levels, and panel (h) shows the 2048 near-surface and volumetric query points used for decoder supervision.

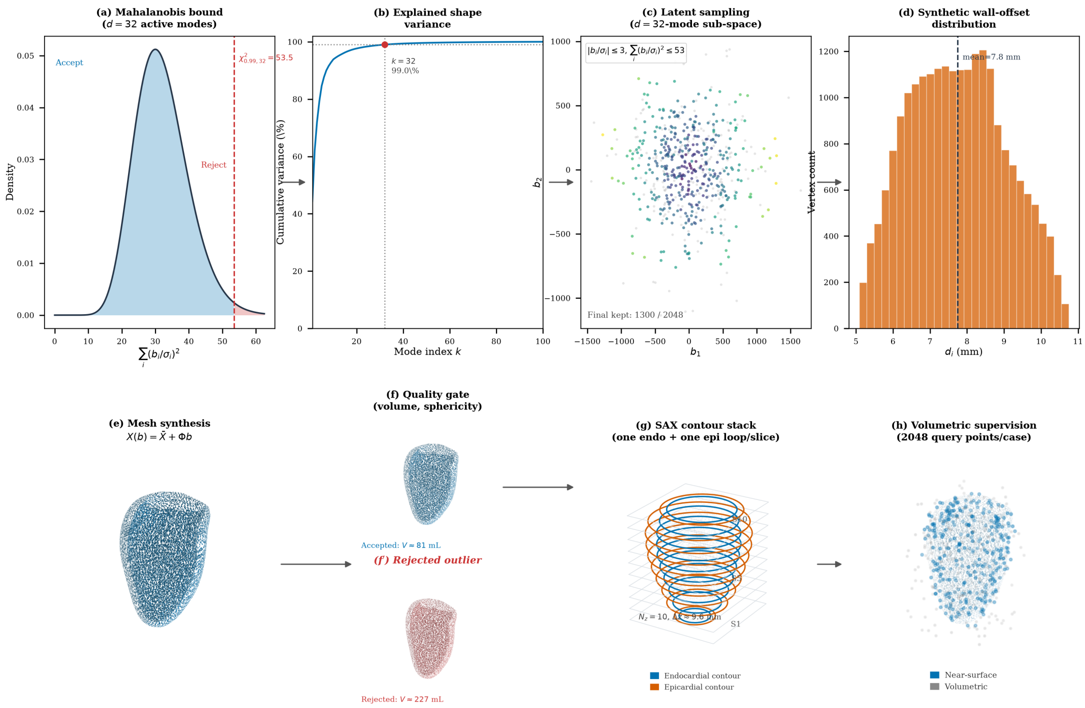

Figure 3.3. Synthetic ED dataset generation from SSM sampling to geometric supervision. Top row: statistical constraints; bottom row: geometric construction and supervision targets.

These fixed SSM-sampling and supervision settings were kept constant across all synthetic ED cases to ensure reproducible pre-training before fine-tuning on real ED and ES data.

3.2.4. Real ED and ES labels: from SAX segmentation to 3D supervision

SSM-derived meshes alone cannot capture the full diversity of clinical acquisition conditions: scanner differences, segmentation quality, image resolution, and the distinct geometry of the end-systolic phase. A second data stream was therefore created by reconstructing watertight three-dimensional LV meshes directly from the segmentation masks provided in the ACDC (Bernard et al., 2018), M&Ms (Campello et al., 2021), and M&Ms-2 (Martín-Isla et al., 2023) datasets, taking both the ED and ES frames so that the model is exposed to both cardiac phases.

These real ED and ES caches are used in the second optimisation stage, after the initial SSM-based training. The reason for this fine-tuning stage is that the synthetic corpus teaches a strong global LV-shape prior, but it does not fully reproduce clinical variability in contour quality, inter-dataset label style, and phase-specific contraction patterns. Fine-tuning on real ED and ES therefore reduces domain shift between synthetic supervision and the target clinical setting while preserving the geometric regularity learned from the SSM.

Operationally, the fine-tuning stage keeps the same model architecture and the same multi-term loss from Section 3.4, but replaces the synthetic-only stream with real-derived cache samples from ED and ES frames. Because the real caches follow the same schema (contours, tissue labels, phase label, mesh supervision, and query points), the training loop can continue from the pre-trained checkpoint without changing model interfaces. The phase label then provides the conditioning signal that allows one shared model to adapt from diastolic to systolic geometry during continued training.

Before describing how these meshes are built, it is useful to look at the raw clinical input, because it makes clear why a reconstruction step is needed at all. Figure 3.4 shows representative SAX slices from the ACDC dataset (Bernard et al., 2018), with the LV cavity, myocardium, and right-ventricular contours overlaid. Each frame is only a two-dimensional cross-section of the heart, and the segmentation is defined one slice at a time. Panel (b) of Figure 3.4 stacks the segmented ED slices at their physical through-plane positions and exposes the central difficulty of the problem: a clinical acquisition provides only a small number of parallel SAX slices separated by several millimetres, so the observed geometry is a set of sparse contour rings rather than a continuous surface.

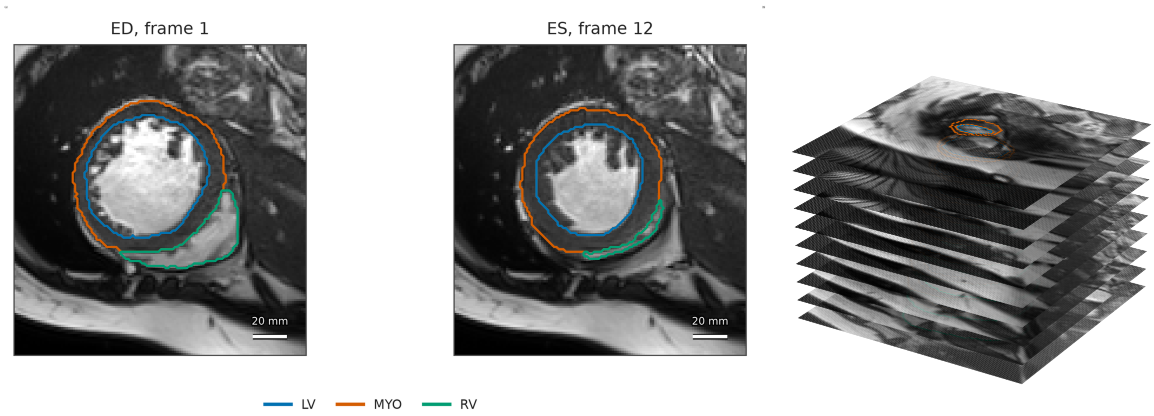

Figure 3.4. Clinical-input view used in the real-data pipeline: (a) representative ACDC ED/ES SAX slices with labels and (b) the corresponding sparse ED SAX stack in physical three-dimensional spacing.

The reconstruction pipeline closes this gap by converting each sparse label stack into a derived, watertight three-dimensional mesh. These meshes are *derived supervision*, not manually annotated 3D ground truth: their geometry is determined by the observed labels and by the explicitly stated resampling, smoothing, and capping operations. This distinction is important when interpreting the later reconstruction results. An earlier ES pipeline attempted to fit the UK Biobank SSM to systolic contours; this was abandoned because forcing a diastolic statistical prior onto systolic geometry produced anatomically unrealistic surfaces with collapsed walls. The direct reconstruction approach instead retains the patient-specific geometry present in the ED or ES labels. The two parallel endocardial and epicardial paths are shown in Figure 3.5; the six cache-construction steps are described below.

Step 1: NIfTI loading and RAS+ canonicalisation. Each segmentation volume is loaded from NIfTI format and reoriented to RAS+ canonical orientation using its affine. This places voxel axes in a consistent patient-coordinate convention while preserving physical spacing. It does not invent a new long axis; the acquired SAX stack supplies the through-plane direction, and an apex–base check is applied later before basal capping.

Step 2: Label harmonisation and binary mask construction. The ACDC, M&Ms, and M&Ms-2 datasets use compatible but not identical label encodings for the LV cavity, myocardium, and right ventricle. Each source is mapped to a common convention before the binary masks are formed. The endocardial mask is defined from the LV cavity label, and the epicardial mask from the union of LV cavity and myocardium labels:

$$M_{\text{endo}} = \mathbb{1}[L = L_{\text{LV}}], \qquad M_{\text{epi}} = \mathbb{1}[L \in \{L_{\text{LV}}, L_{\text{MYO}}\}], \tag{3.4}$$

where L is the label image, L_{LV} is the LV cavity label, and L_{MYO} is the myocardium label.

Step 3: Isotropic resampling and label cleaning. The segmentation volume is resampled to a target spacing of 1.5 mm in all three dimensions using nearest-neighbour interpolation to avoid label blending. Within each SAX slice, the largest connected component of each label is retained and any holes are filled. This removes small disconnected label fragments that can arise near the basal and apical boundaries of the stack. The largest three-dimensional component is then retained for each mask, and the endocardial mask is unioned into the epicardial mask to enforce $M_{\mathrm{endo}} \subseteq M_{\mathrm{epi}}$. The apex is placed at low z and the base at high z using the relative cross-sectional areas at the two ends of the stack. This establishes a common direction for the basal operation without assuming that RAS+ alone defines cardiac anatomy.

Step 4: Marching cubes surface extraction. After three-dimensional component filtering, the binary mask is Gaussian-smoothed at a scale of 1.5 mm and its 0.5 isosurface is extracted with marching cubes (Lorensen & Cline, 1987). Smoothing before isosurface extraction suppresses the voxel staircase that would result from applying marching cubes directly to discrete labels. Panels (d) and (d′) of Figure 3.5 provide an unsmoothed voxel-surface visualisation of this issue for the endocardial and epicardial masks, respectively.

Step 5: Post-processing—smoothing, basal cap, and decimation. The raw marching-cubes surface is processed in three sub-steps. Taubin λ/μ smoothing (Taubin, 1995) (25 iterations, $\lambda = 0.53$, $\mu = -0.55$) reduces residual staircase artefacts while approximately preserving global volume. Because the label stack ends at the valve plane, marching cubes can create a rounded, non-anatomical basal dome. The upper 6% of the apex–base extent is therefore removed and the resulting opening is capped by a plane perpendicular to the established long-axis direction. Cotangent-Laplacian fairing (3 iterations) then polishes the cap join, and quadric-error decimation (Garland & Heckbert, 1997) limits the mesh to at most 4000 vertices. Panels (e) and (e′) of Figure 3.5 show the Taubin-smoothing effect; the cache meshes receive the complete post-processing sequence described here. The basal-capping operation removes the rounded dome that marching cubes creates at the valve plane using a plane perpendicular to the long axis, replacing it with an anatomically flat cap.

Step 6: Anatomical quality checks and contour extraction. Before a mesh is accepted into the training cache, seven anatomical prechecks are applied. These checks address a fundamental challenge in deriving 3D meshes from sparse segmentation masks: the marching-cubes algorithm will faithfully reconstruct *whatever surface is implied by the voxel labels*, regardless of whether that surface is anatomically plausible. When a label stack is incomplete, corrupted by noise, or represents a long-axis rather than short-axis acquisition, the resulting mesh may be geometrically valid yet anatomically meaningless. Each of the seven criteria below targets a distinct failure mode. A case that fails any single check is excluded from the training cache, because training on implausible geometry would degrade the model's ability to learn anatomically consistent LV shape priors.

Apical taper. The cross-sectional area of the endocardial contour must decrease monotonically from the mid-cavity toward the apex. This criterion enforces the natural conical taper of the LV, where the apical region is narrower than the basal region. Cases where the apical end is wider than the mid-cavity are rejected, as this pattern indicates either an inverted coordinate assignment (base and apex swapped) or a truncated stack that does not extend to the true apex. Either case would inject contradictory geometric supervision into training.

Basal support. The most basal slice must have a cross-sectional area consistent with the valve-plane region. Very small or nearly absent basal contours suggest that acquisition did not reach the anatomical base, leaving the basal boundary weakly constrained. In that situation, the basal-capping operation described in Step 5 would close the mesh at an arbitrary location rather than at a physiologically meaningful plane, creating a derived supervision artefact.

Centreline drift. The lateral displacement of slice-wise centroids along the long axis is bounded to detect discontinuous contour shifts between adjacent slices. Excessive drift is typically associated with segmentation inconsistency, motion effects, or severe orientation mismatch. When unfiltered, this failure mode produces kinked or twisted reconstructions that are geometrically connected but anatomically implausible as LV surfaces.

Contour regularity. Within each SAX slice, the sampled endocardial and epicardial rings must remain smooth and approximately convex. Jagged boundaries, abrupt concavities, or self-intersections are common signs of noisy labels or local segmentation failure. Although smoothing can attenuate high-frequency artefacts, it cannot reliably correct topologically degenerate contours. Such cases are therefore excluded before contour caching.

Endocardium-inside-epicardium ordering. At every axial level, the endocardial contour must lie inside the epicardial contour with positive separation. This preserves the basic anatomical relation between cavity and myocardial wall. Cases with inverted ordering, overlap, or crossing loops are removed because they correspond to impossible supervision targets for the coupled endocardial–epicardial decoder.

Wall proxy thickness. A slice-wise proximity check verifies that the average endocardial–epicardial separation remains within plausible bounds for the target cohort. Extremely small separation often indicates segmentation bleed-through or label merging, while unusually large separation often reflects annotation leakage into non-myocardial structures. Excluding these outliers stabilises training and keeps the supervision distribution consistent with intended LV anatomy.

Long-axis ratio. The ratio of apex-to-base extent to maximum short-axis diameter is constrained to reject global shape outliers. Very low ratios suggest severely distorted elongated geometries or axis-definition errors, whereas very high ratios are commonly associated with basal label fragments or implausible cavity dilation in the reconstructed mask volume. Constraining this ratio helps preserve a coherent global LV aspect profile across the retained cache.

Cases that fail any of these checks are discarded from the cache. Accepted meshes are centred and normalised jointly, then intersected with 10 equally spaced axial planes after a 5% apical and basal margin. The largest intersection loop at each plane is resampled to 60 points for each available tissue. These are the contour points observed by the encoder. Separately, 2048 near-surface and volumetric query points are sampled and labelled by mesh occupancy for decoder supervision. Thus, the 2D-looking encoder input and the volumetric target are two representations of the same derived mesh, rather than independent annotations.

Figure 3.5 provides a visual summary of the conversion from clinical labels to mesh-ready supervision.

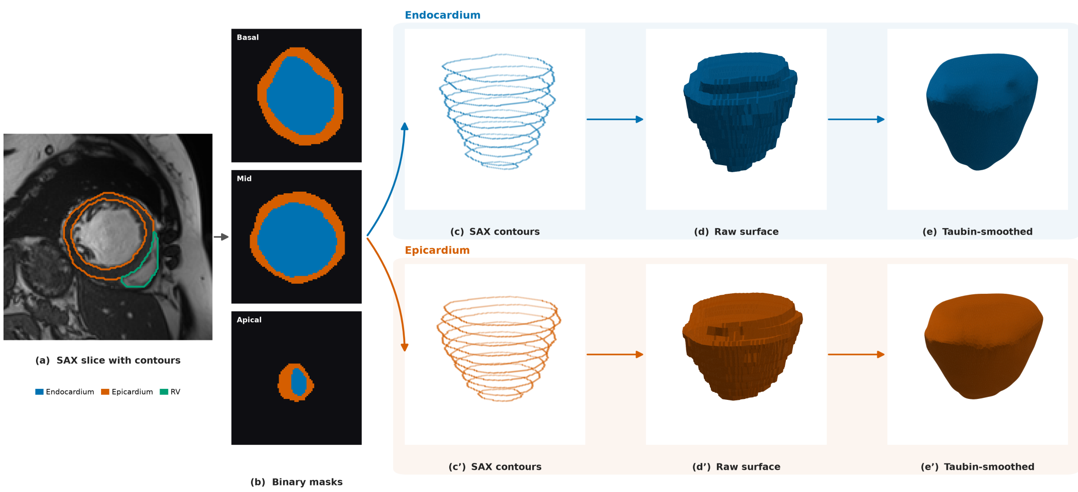

Figure 3.5. Real ED clinical-label-to-mesh pipeline (ACDC): from SAX labels and binary masks to endocardial and epicardial surface branches used for derived supervision.

Panel (a) shows the acquired SAX image with LV cavity, myocardium, and RV labels, and panel (b) shows the corresponding binary endocardial and epicardial masks after label harmonisation. The pipeline then branches into separate endocardial and epicardial paths: panels (c)–(e) and panels (c′)–(e′), respectively.

In both branches, contours are placed in physical three-dimensional positions, a voxel surface view makes discretisation artefacts explicit, and Taubin smoothing illustrates their reduction before later post-processing. The figure is intentionally a process overview; the full cache-construction sequence also includes basal capping, cap fairing, decimation, anatomical prechecks, and re-slicing as described in Steps 5–6.

Table 3.1. Main operations in the real ED/ES mesh-preparation pipeline.

Operation	Purpose
RAS+ canonicalisation	Standardises patient-coordinate orientation while preserving physical spacing
Label harmonisation	Converts ACDC, M&Ms, and M&Ms-2 to a common LV/myocardium mask
Isotropic resampling at 1.5 mm	Reduces anisotropy before 3D surface extraction
Per-slice cleaning	Removes small 2D fragments and fills holes
3D largest-component filtering	Keeps the main anatomical LV region
Gaussian mask smoothing and marching cubes	Suppresses staircase artefacts and extracts the surface
Taubin smoothing and Laplacian fairing	Reduces residual artefacts while preserving global shape
Basal cap perpendicular to the apex–base axis	Replaces the non-anatomical basal dome at the valve-plane boundary
Quadric decimation to 4000 vertices	Produces a consistently sized mesh for all cases
Anatomical prechecks (7 criteria)	Rejects geometrically implausible or corrupted cases

3.2.5. Shared cache representation and contour augmentation

All cases are stored in a common cache representation so that the model receives the same interface regardless of whether a sample originates from the SSM or from a clinical segmentation. A cache item combines the sparse contour observation with mesh-derived supervision and the metadata needed to keep both representations in a shared coordinate frame. The ED and ES caches contain 10 SAX levels, 60 points per available contour ring, and 2048 query points per case. This design deliberately separates what the model observes from what it is allowed to learn from: the encoder receives only contour coordinates, tissue identity, and cardiac phase, whereas the decoder is supervised by surface and volumetric samples derived from the corresponding mesh. Thus, at inference time, the same model can reconstruct geometry from sparse contour observations alone.

Table 3.2 makes this separation explicit. The first three fields describe the clinical-style observation supplied to the encoder; the remaining fields are available only during training, evaluation, or visualisation. Keeping these roles distinct prevents mesh-derived information from leaking into the encoder input while allowing dense supervision of the implicit field.

Table 3.2. Fields of one shared cache item and their role in training.

Field	Origin	Role
Contour points	Synthetic mesh slicing or clinical mask re-slicing	Sparse SAX observation for the encoder and contour-anchor loss
Tissue label	Contour extraction	Distinguishes endocardial from epicardial points during tissue-specific pooling
Phase label	Dataset metadata	Conditions the encoder on ED or ES geometry
Surface samples	Prepared endocardial and epicardial meshes	Surface-fitting and normal-alignment supervision
Query points and targets	Near-surface and volumetric mesh sampling	Dense SDF, occupancy, sign, and exterior supervision
Mesh metadata	Preprocessing pipeline	Coordinates, scale, faces, and orientation for evaluation and visualisation

This common contract enables two-stage optimisation without dataset-specific changes to the architecture or loss. The model is first trained on synthetic ED samples, which establish a broad geometric prior, and is then fine-tuned on real ED and ES samples, which introduce patient-specific anatomy and phase-dependent shape variation. The loader changes only the source stream between stages; the fields consumed by the training loop remain unchanged.

Contour augmentation is applied only to the synthetic ED training observations, after the train/validation/test split. The target mesh, surface samples, query points, occupancy labels, tissue labels, and phase label remain fixed. This is therefore an observation-robustness intervention rather than the creation of a geometrically transformed anatomical target. Such label-preserving perturbations are commonly used to improve robustness when labelled data are limited (Chlap et al., 2021; Shorten & Khoshgoftaar, 2019).

For each augmented synthetic sample, the loader draws small independent in-plane translations per slice, point-wise in-plane jitter, global scale jitter, and contour-point dropout. Slice dropout and long-axis rotation are applied probabilistically while retaining at least three slice levels. The endocardial and epicardial contours within a slice receive the same translation, preserving their local relationship. As illustrated in Figure 3.6, these perturbations ask the encoder to recover one fixed mesh-derived target from plausible variations of its sparse SAX observation.

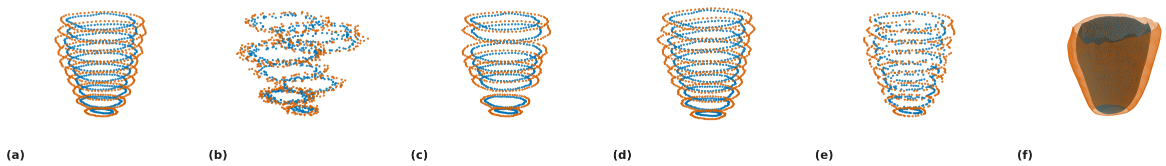

Figure 3.6. Examples of contour-input augmentation. Endocardial and epicardial points are shown in blue and orange, respectively.

In Figure 3.6, panel (a) shows the unaltered contour stack, while panels (b), (d), and (e) show slice translation, slice rotation, and contour-point dropout, respectively. Panel (f) shows the corresponding 3D mesh label. Thus, the sparse SAX observation changes across the augmented examples, whereas the derived mesh and all decoder-supervision targets remain fixed; the figure illustrates robustness of the encoder input rather than transformation of the anatomical ground-truth geometry.

3.2.6. Dataset composition and splits

This subsection reports how many samples each data stream finally contributes and how the real streams are divided for training and evaluation. The real data come from two public collections: the Automated Cardiac Diagnosis Challenge (ACDC) and the second Multi-Centre, Multi-Vendor and Multi-Disease Cardiac Segmentation challenge (M&Ms-2). Together they provide 460 patients (100 from ACDC and 360 from M&Ms-2). For each patient we reconstruct one mesh at end-diastole (ED, the fully filled ventricle) and one at end-systole (ES, the contracted ventricle), giving two real streams. The synthetic stream is generated from the UK

Digital Heart Project SSM: of its 100 modes, the first 32 already capture 99% of the shape variance, so we sample within that sub-space. Drawing shape coefficients under the Mahalanobis bound of Equation (3.2) and rejecting implausible shapes yields 1300 accepted meshes out of 2048 candidates (63%).

Not every real patient produces a usable mesh. A segmentation may be a long-axis view, cover too few slices, or fail an anatomical plausibility check, such as an endocardium that is not fully enclosed by the epicardium. After these quality gates, 444 of the 460 patients yield a valid ED mesh and 423 yield a valid ES mesh. Figure 3.7 summarises this attrition for the three streams.

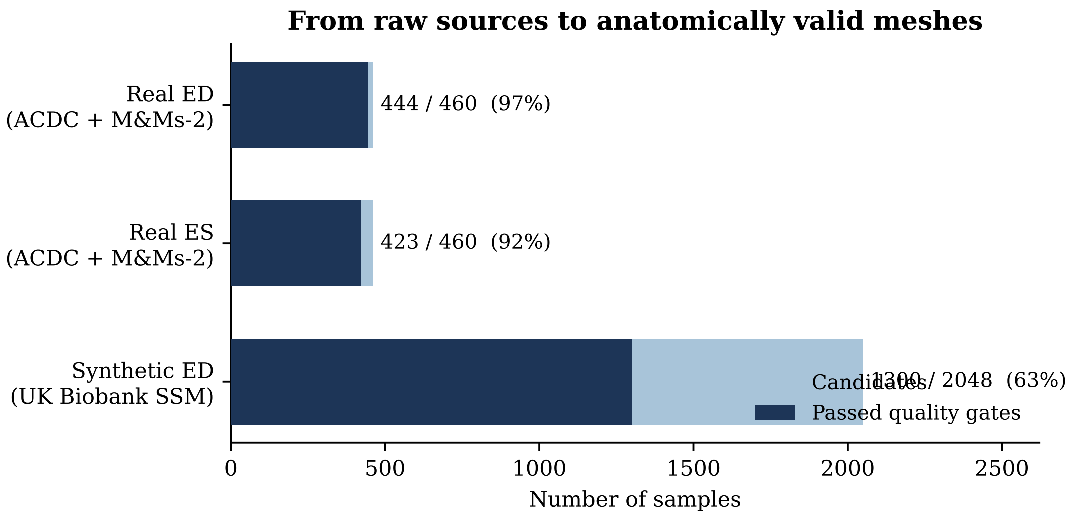

Figure 3.7. Number of candidate samples and how many survive the quality gates for each data stream. The real streams start from the 460 indexed patients; the synthetic stream starts from the 2048 sampled shape vectors.

To prevent information from leaking between training and evaluation, the real streams are split by patient: all meshes from a given patient stay within a single split. Figure 3.8 shows the resulting composition. The ED stream is divided into 310 training, 67 validation, and 67 test patients, and the ES stream into 338 training, 66 validation, and 19 test patients. The synthetic ED meshes are used only to enrich the pre-training stage and are never placed in the validation or test sets.

Table 3.3 collects the final sample counts. In total, the model is trained and evaluated on 867 real LV meshes, supported by up to 1300 synthetic ED meshes during pre-training.

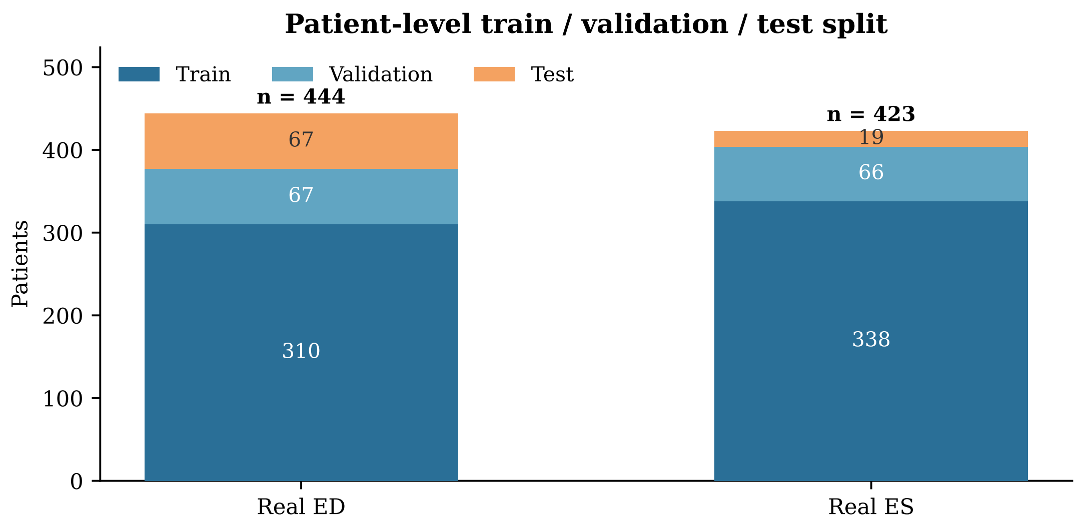

Figure 3.8. Patient-level train, validation, and test split for the two real streams. Splitting by patient guarantees that no patient appears in more than one partition.

Table 3.3. Final sample counts for the three data streams. Real streams report patients that pass the quality gates; the synthetic stream reports accepted meshes.

Stream	Source	Phase	Valid	Split		
			meshes	Train	Val.	Test
Real ED	ACDC + M&Ms-2	ED	444	310	67	67
Real ES	ACDC + M&Ms-2	ES	423	338	66	19
Synthetic ED	UK Biobank SSM	ED	1300	pre-training only		

3.3. Model Architecture

The model has two parts: an encoder and a decoder. The encoder reads the sparse contour points and summarises them into a single latent vector that describes the LV geometry of one case. The decoder takes that vector together with a 3D query point and returns how far the point is from the endocardial surface. Because the decoder can be evaluated at any point in space, the reconstructed surface is not tied to a fixed grid or a fixed number of vertices, and a mesh is produced only at the end by sampling the decoder densely. The encoder follows the point-cloud principle of Qi et al. (2017) and the implicit decoder follows the signed-distance idea of Park et al. (2019).

This split is a natural fit for the problem, because the input and the output have very different structure. The input is a small, unordered set of slice points, so a point-cloud encoder is appropriate: it does not require the points to be ordered or arranged on a grid. The output is a smooth, continuous surface, so an implicit decoder is appropriate: it can be sampled at any resolution and turned into a mesh only when needed. The key addition in this work is that the epicardium is not predicted on its own. Instead, it is defined as an outward offset from the endocardium, which forces the wall to have a positive thickness everywhere.

3.3.1. Contour encoder

The encoder receives, for each contour point, its 3D coordinates together with a label indicating whether the point is endocardial or epicardial and a label indicating the cardiac phase. A small shared network processes every point independently and turns it into a feature vector. The features of the endocardial points and the features of the epicardial points are then combined separately by taking, for each feature, its maximum over the points. This max operation is what makes the encoder independent of the order in which the points are listed, which is important because contour points have no natural ordering across patients or slices. The two resulting descriptors are joined and projected into a single 256-dimensional latent vector z.

More formally, let $P = \{p_i\}_{i=1}^{N}$ be the contour points of one case, where each p_i carries coordinates, a tissue label, and a phase label. The shared point network computes

$$h_i = \phi_{\text{enc}}(p_i), \tag{3.5}$$

where ϕ_{enc} is the shared network. The endocardial and epicardial descriptors are then obtained by max-pooling within each tissue,

$$z_{\text{endo}} = \max_{i \in \mathscr{E}} h_i, \qquad z_{\text{epi}} = \max_{i \in \mathscr{P}} h_i, \tag{3.6}$$

where $\mathscr{E}$ and $\mathscr{P}$ are the endocardial and epicardial point sets, and the final latent code is

$$z = W_z[z_{\text{endo}}, z_{\text{epi}}] + b_z. \tag{3.7}$$

Pooling the two tissues separately keeps their information distinct. The endocardial points describe the shape of the cavity, while the epicardial points describe the outer boundary of the wall. If all points were pooled together, these two roles would be mixed and the decoder would have a harder time telling the cavity apart from the wall. Keeping them separate gives the latent code explicit access to both surfaces before they are combined.

3.3.2. Implicit decoder and positive-wall coupling

Before a query point is given to the decoder, its coordinates are expanded with a Fourier positional encoding. This simply adds a few sine and cosine copies of the coordinates at different frequencies, which helps the network represent fine geometric detail instead of only smooth, blurry shapes (Tancik et al., 2020). Three frequency bands are used. For a query point $x \in \mathbb{R}^3$ the encoding is

$$\gamma(x) = [x, \sin(2^0\pi x), \cos(2^0\pi x), \ldots, \sin(2^{L-1}\pi x), \cos(2^{L-1}\pi x)], \tag{3.8}$$

with $L = 3$. The decoder itself is an eight-layer fully connected network of width 512 with a skip connection at the middle layer. It uses a smooth activation (softplus) so that the reconstructed surface is smooth and free of the sharp creases that a ReLU activation would introduce.

The decoder produces two outputs. The first is the signed distance to the endocardium, f_{endo}. The second is a positive offset δ, which represents the local wall thickness, that is, how far the epicardium lies outside the endocardium at that point. The offset is squeezed into a fixed positive range by a sigmoid,

$$\delta(x,z) = \tau_{\min} + (\delta_{\text{cap}} - \tau_{\min})\sigma\left(g_\delta(x,z)\right), \tag{3.9}$$

where g_δ is the raw decoder output, σ is the sigmoid, and $\tau_{\min}$ and δ_{cap} are the smallest and largest allowed wall thickness. The epicardial field is then

$$f_{\text{epi}}(x,z) = f_{\text{endo}}(x,z) - \delta(x,z). \tag{3.10}$$

Because Equation (3.9) always keeps $\delta \geq \tau_{\min} > 0$, the epicardial surface can never touch or cross the endocardial surface: the two are always separated by at least $\tau_{\min}$. This is the mechanism that guarantees a positive wall thickness everywhere, rather than hoping the network learns it from data. In the final configuration $\tau_{\min} = 0.05$ and $\delta_{\text{cap}} = 0.45$ in normalised units, which corresponds to roughly 1.25–11.25 mm once mapped back to millimetres.

Table 3.4. Main architecture settings of the model.

Component	Configuration
Input features	3D coordinates, tissue label, and cardiac phase
Encoder type	Shared network with endocardial and epicardial max-pooling
Latent dimension	256
Positional encoding	Fourier features with $L = 3$ frequency bands
Decoder	8-layer network, width 512, skip connection at layer 4
Activation	Softplus (smooth)
Output heads	Endocardial signed distance and positive wall-thickness offset
Thickness range	$\tau_{\min} = 0.05$, $\delta_{\text{cap}} = 0.45$ in normalised units
Inference grid	96^3 query grid for marching cubes

3.4. Training and Inference

This section explains how the model is trained from the shared cache and how a surface mesh and a wall-thickness map are produced once training is finished. Training proceeds in two stages: the model is first pre-trained on the synthetic SSM cases and then fine-tuned on the real ED and ES cases. In both stages it must learn two things together: a valid distance field for the endocardium, and a sensible relationship between the endocardium and the epicardium.

3.4.1. Training data mix and augmentation

Training follows the two-stage strategy introduced in Section 3.2. In the first stage, the model is pre-trained only on the synthetic ED cases generated from the SSM. This large, controlled corpus teaches a strong prior for LV shape and tolerates aggressive augmentation. In the second stage, the pre-trained model is fine-tuned on the real cases reconstructed from ACDC, M&Ms, and M&Ms-2, covering both the ED and ES phases. Fine-tuning starts from the pre-trained weights and continues with the same architecture and loss, so the model keeps the geometric regularity learned from the SSM while adapting to real segmentation quality and to the systolic phase. Throughout both stages the phase label is provided as input, so a single model handles ED and ES, and augmentation is applied only to the contour input; the target meshes and the cached supervision are never changed.

The two stages are complementary. The synthetic ED cases give a large set of anatomically valid shapes but do not reproduce every artefact seen in real segmentations, which is why they are used for pre-training rather than as the final training data. The real ED and ES cases then supply genuine patient anatomy from several datasets and let the model learn how the shape changes between the two phases. To avoid information leaking between the training, validation, and test sets, the real cases are split by patient before any augmentation is applied. Because the augmentation only perturbs the observed contours, each real target stays tied to the anatomy of its own patient.

In practice, this means the encoder is trained to be robust to imperfect, noisy inputs, such as small shifts, jitter, missing slices, or dropped points, while the decoder is always supervised against a fixed, consistent 3D geometry.

3.4.2. Loss function

The training loss is a weighted sum of several terms, each responsible for one aspect of a good reconstruction. Some terms pull the predicted surfaces onto the target meshes, some keep the distance field well-behaved, some keep the surface faithful to the observed slices, and some prevent the wall from collapsing. The full objective is

$$\begin{aligned}\mathcal{L}_{\text{total}} = {} & \lambda_{\text{surf}}\mathcal{L}_{\text{surf}} + \lambda_{\text{eik}}\mathcal{L}_{\text{eik}} + \lambda_{\text{off}}\mathcal{L}_{\text{off}} + \lambda_{\text{normal}}\mathcal{L}_{\text{normal}} \\ & + \lambda_{\text{WT}}\mathcal{L}_{\text{WT}} + \lambda_{\text{L1}}\mathcal{L}_{\text{L1}} + \lambda_{\text{anchor}}\mathcal{L}_{\text{anchor}} + \lambda_{\text{sign}}\mathcal{L}_{\text{sign}} \\ & + \lambda_{\text{extsign}}\mathcal{L}_{\text{extsign}} + \lambda_{\text{bbox}}\mathcal{L}_{\text{bbox}} + \lambda_z\|z\|_2^2. \end{aligned} \tag{3.11}$$

The most important terms are the surface term, which places the two surfaces on their target meshes, and the eikonal term, which keeps the output a proper signed distance field by encouraging its gradient to have unit length (Gropp et al., 2020). The wall-thickness term reinforces the built-in lower bound on δ by discouraging the wall from ever thinning toward its minimum. The remaining terms handle finer effects such as surface orientation, agreement with the input contours, and the suppression of spurious surfaces outside the heart; their individual roles are listed in Table 3.5. The eikonal term is computed in full precision to keep its gradient calculation numerically stable.

Table 3.5. Purpose of the main training-loss terms.

Term	Purpose	Effect on reconstruction
$\mathscr{L}_{\text{surf}}$	Fits zero-level sets to surface samples	Places endocardial and epicardial surfaces on the target meshes
$\mathscr{L}_{\text{eik}}$	Encourages $\|\nabla f\| = 1$	Regularises the decoder as a signed-distance field
$\mathscr{L}_{\text{off}}$	Repels off-surface points from zero	Reduces spurious zero crossings away from the myocardium
$\mathscr{L}_{\text{normal}}$	Aligns predicted and target normals	Improves local surface orientation
$\mathscr{L}_{\text{WT}}$	Enforces the wall-thickness floor	Discourages collapse toward an unrealistically thin wall
$\mathscr{L}_{\text{L1}}$	Supervises cached query-point distances	Provides dense volumetric SDF information
$\mathscr{L}_{\text{anchor}}$	Anchors input contour points to the surface	Preserves agreement with observed SAX contours
Sign terms	Enforce inside/outside consistency	Stabilise the relative position of endocardium and epicardium
Bbox term	Pushes points outside the contour box outward	Suppresses exterior false surfaces

3.4.3. Optimisation and hardware

Both the SSM pre-training stage and the real-data fine-tuning stage use the same optimisation setup: the AdamW optimiser at a learning rate of 2×10^{-5} and weight decay 5×10^{-4}, with the learning rate following a cosine schedule and early stopping ending each stage once the validation loss stops improving. Fine-tuning simply resumes from the pre-trained weights rather than starting from scratch. To fit the model in the available GPU memory, training uses mixed precision, gradient clipping at 1.0 for stability, and gradient accumulation to reach a larger effective batch size. Each stage runs for at most 400 epochs, though early stopping usually ends it sooner. Training was carried out on two GPUs in parallel. A light weight-clipping strategy was used instead of the more expensive spectral normalisation, since the decoder is evaluated at many points per step and GPU memory is the main constraint.

Table 3.6. Training configuration used for the model.

Setting	Value
Training strategy	SSM pre-training, then fine-tuning on real ED/ES
Training mode	Combined ED and ES with phase conditioning
Augmentation	Contour-input perturbations for synthetic and real cases
Optimiser	AdamW
Learning rate	2×10^{-5}
Weight decay	5×10^{-4}
Maximum epochs	400 per stage
Batch size	8, with gradient accumulation
Gradient clipping	1.0
Precision	Automatic mixed precision
Hardware	2 GPUs in parallel
Inference grid	96^3

3.4.4. Inference and analytic wall-thickness extraction

At inference time, the model turns the input contours into the latent vector z and then evaluates the decoder on a regular 96^3 grid of points. Marching cubes reads the zero-level sets of f_{endo} and f_{epi} from this grid and produces the endocardial and epicardial meshes (Lorensen & Cline, 1987). No mesh repair or hole filling is applied afterwards; the surfaces come directly out of the model. A wall-thickness value is available for free at every endocardial vertex, because the decoder already predicts the offset δ there; converting it to millimetres gives

$$t_{\text{model}}(x) = s\,\delta(x,z), \tag{3.12}$$

where s is the scale that maps normalised coordinates back to millimetres. This thickness is not measured after the fact by comparing two meshes; it is a direct output of the model. The selected measurement algorithms in Section 3.5 are therefore used as independent points of comparison on the same geometry, not as the only way to obtain thickness.

3.5. Wall-Thickness Evaluation Protocol

The wall-thickness evaluation is performed on the model-generated geometry. The segmentation volume is used to extract contour rings and to orient the AHA regional coordinate system, but the measured surfaces are the endocardial and epicardial meshes produced by the trained model. This distinction is important: the evaluation tests thickness estimation on the reconstructed implicit model, not directly on raw segmentation meshes. Myocardial wall thickness is itself a routine indicator of cardiac health, because a wall that is abnormally thin can signal infarction or scarring, while an abnormally thick wall can signal hypertrophy. The evaluation is therefore organised in three steps: first the algorithms that measure thickness on the geometry, then the resulting local three-dimensional thickness field, and finally its aggregation into the clinical AHA-17 regional summary.

3.5.1. Selected thickness methods

We evaluate four wall-thickness algorithms on the reconstructed geometry, chosen to span the accuracy–robustness trade-off relevant to a signed-distance reconstruction across the main methodological families — volumetric distance fields, ray intersections, and partial-differential-equation (PDE) formulations. Three are used as the primary estimators: the Laplace field method as the established transmural reference, the Yezzi–Prince Eulerian PDE method as a more recent and numerically more stable formulation, and the signed-distance cone-ray method as a promising surface-based estimator that aligns naturally with the SDF geometry produced by the model. The Euclidean Distance Transform (EDT) boundary sum is retained as a volumetric baseline for context. Each method returns local thickness values in millimetres at endocardial vertices or projected endocardial locations. The outputs are summarised with mean, median, standard deviation, fifth percentile, ninety-fifth percentile, finite-value fraction, and runtime, and are visualised as 3D colour maps and AHA-17 bullseye plots.

Table 3.7. Wall-thickness methods selected for evaluation on the model-generated LV geometry, with the Euclidean Distance Transform boundary sum kept only as a baseline.

Method	Category	Role	Main measurement principle
Laplace field	PDE	State of the art	Transmural field lines from a harmonic potential
Yezzi–Prince	PDE	Recent	Eulerian transport equations along the transmural field
SDF cone rays	Ray / SDF	Promising	Median of multiple cone rays around the surface normal
EDT boundary sum	Volumetric	Baseline	Sum of endocardial and epicardial distance transforms

As a volumetric baseline, the EDT boundary sum computes, at each myocardial voxel x, the sum of the distances to the endocardial and epicardial surfaces,

$$t^{\mathrm{EDT}}(x) = D_{\mathrm{endo}}(x) + D_{\mathrm{epi}}(x), \tag{3.13}$$

where D_{endo} and D_{epi} are the Euclidean distance transforms of the two boundaries. This is exact for planar parallel walls but is limited by voxel resolution and slightly overestimates thickness in strongly curved regions, which motivates the three selected methods below.

The Laplace field method is the established transmural reference. It defines a smooth scalar field ψ over the myocardium by solving

$$\nabla^2 \psi = 0, \qquad \psi|_{\mathrm{endo}} = 0, \qquad \psi|_{\mathrm{epi}} = 1. \tag{3.14}$$

The local wall thickness is then estimated from the gradient magnitude along the transmural field. This family of methods is attractive because harmonic field lines do not cross, but numerical errors can appear where the wall is thin or where the field gradient is small (Jones et al., 2000). The more recent Yezzi–Prince approach addresses this by solving Eulerian transport equations along the same transmural field and combining the endocardial and epicardial travel times (Yezzi & Prince, 2003). In simplified form,

$$\nabla \psi \cdot \nabla u = -1, \qquad t = u + v, \tag{3.15}$$

where u and v are the propagation distances from opposite boundaries.

The cone-ray method uses the endocardial surface normal but reduces sensitivity to a single noisy ray. Inspired by the shape diameter function (Shapira et al., 2008), for each endocardial point several rays are cast inside a cone around the normal direction, and the median valid hit distance is used:

$$t_i^{\mathrm{cone}} = \mathrm{median}_{k=1}^{K}\, d_{ik}, \qquad K = 7. \tag{3.16}$$

In the notebook implementation the cone half-angle is 30°. This makes the estimate more robust than a single normal ray while still preserving an approximately transmural interpretation.

Figure 3.9 contrasts how the four estimators measure thickness on the same wall cross-section. The EDT baseline sums the distances from an interior point to both boundaries; the Laplace method follows the transmural streamlines of a harmonic potential whose nested iso-surfaces are shown as dashed lines; the Yezzi–Prince method integrates two travel times u and v along that same field and adds them; and the cone-ray method casts a fan of rays around the endocardial normal and keeps the median hit distance.

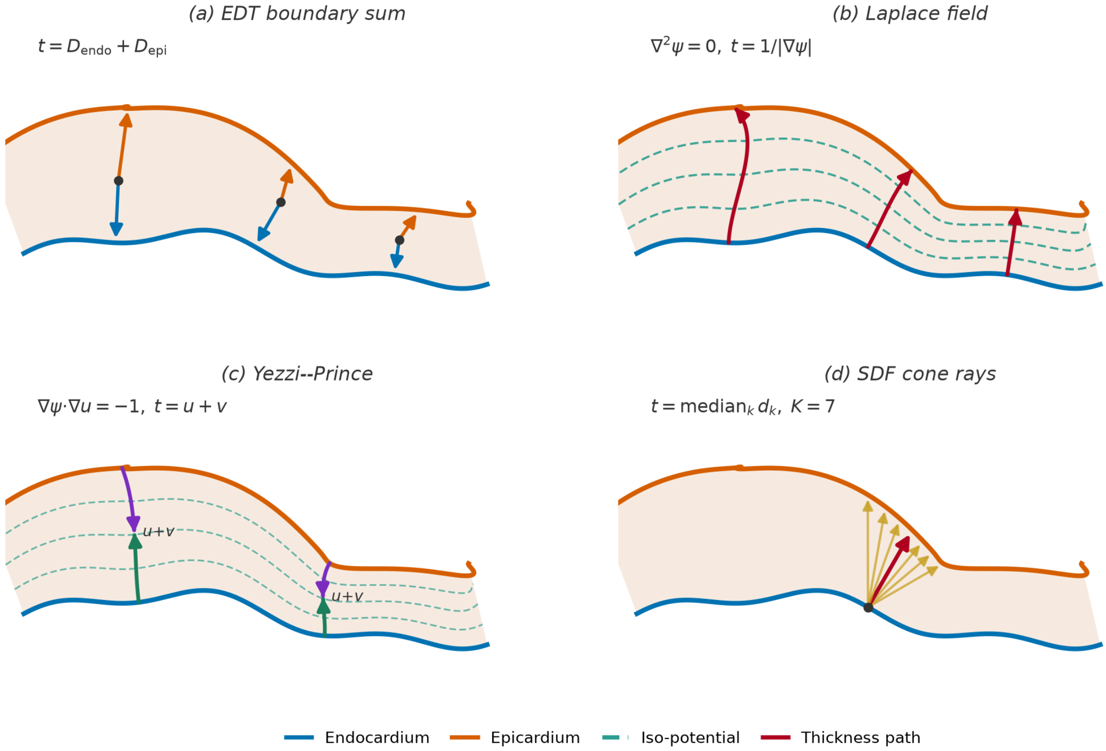

Figure 3.9. Conceptual comparison of the four wall-thickness estimators on a single myocardial cross-section, with the endocardium in blue and the epicardium in orange. (a) The EDT boundary sum adds the distances from an interior point to each boundary. (b) The Laplace field method follows the curved transmural streamlines of a harmonic potential, which cross the iso-potential surfaces (dashed lines) at right angles, and estimates thickness from the gradient magnitude. (c) The Yezzi–Prince method propagates two travel times u and v from opposite boundaries along that same curved field and sums them. (d) The SDF cone-ray method casts a fan of rays within a cone around the endocardial normal and takes the median intersection distance (highlighted), making it robust to any single ill-oriented ray.

3.5.2. Local three-dimensional wall thickness

Each of the selected methods produces, before any regional summary, a continuous thickness field defined over the whole reconstructed geometry. Because the model decoder can be evaluated at any point in space, the endocardial and epicardial surfaces are watertight and densely sampled, and a thickness value is attached to each endocardial vertex. This produces a local three-dimensional wall-thickness map rather than a handful of per-slice numbers, as shown in Figure 3.10, where the LV surface is coloured by local thickness from the thin apical region to the thicker basal wall.

This local map is the primary output of the wall-thickness stage. Presenting the measurement first as a full surface field, and only afterwards as regional segment averages, makes the spatial distribution of thickness explicit: localised thinning or thickening remains visible on the surface even when it would be averaged away inside a single AHA segment. The colour map uses a fixed thickness range across cases so that maps from different patients and cardiac phases can be compared directly.

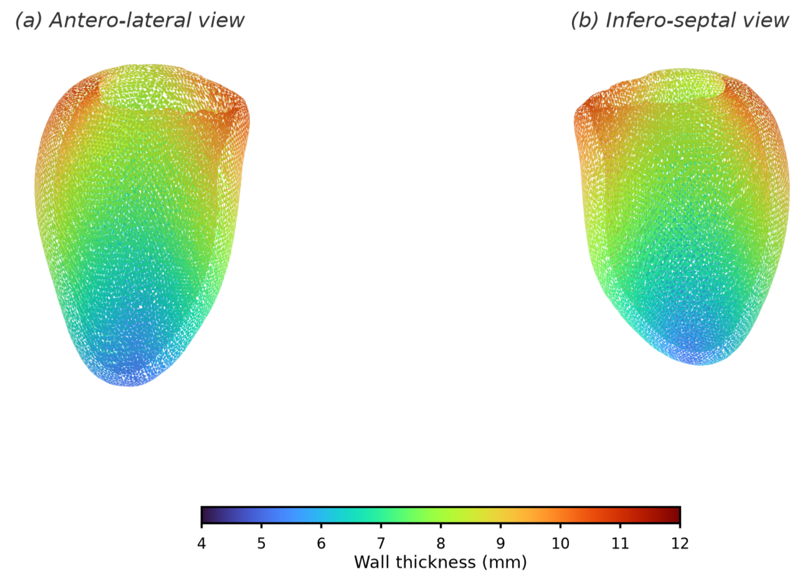

Figure 3.10. Local three-dimensional wall-thickness map on the reconstructed LV endocardial surface, shown from two opposing viewpoints. Each vertex is coloured by its local wall thickness on a fixed millimetre scale, so that the continuous spatial variation from the thin apex to the thicker base is visible directly on the geometry rather than only as regional averages.

3.5.3. Regional AHA-17 mapping

The dense surface field is finally aggregated into the 17-segment model of the American Heart Association (AHA), the accepted clinical language for reporting regional myocardial measurements (Cerqueira et al., 2002). Conventionally the wall thickness in each segment is read semi-manually from a few short-axis slices, which is observer-dependent, coarsely sampled, and sensitive to the choice of slice and measurement point. Here the same segment values are produced objectively from the local field of Section 3.5.2, by cutting the reconstructed surface into the AHA segments and averaging within each one.

The cut is defined in the LV's own anatomical frame, which first requires identifying the long-axis direction, distinguishing base from apex, and fixing a septal reference direction; the wall-thickness notebook includes an anatomical QA step that visualises this base/apex orientation and septal alignment before the segments are formed. The surface is then partitioned in two stages, as illustrated in Figure 3.11. First, along the long axis, the endocardial vertices are split by their normalised apex–base position into a basal ring, a mid-cavity ring, an apical ring, and a small apical cap. Second, within each ring the vertices are divided angularly around the long-axis centreline: the basal and mid rings are cut into six sectors each and the apical ring into four, while the apical cap forms the single seventeenth segment. Numbering the sectors from the anterior–septal reference and unrolling the rings onto concentric circles yields the familiar bullseye, in which the basal ring is the outer annulus, the mid ring the middle annulus, the apical ring the inner annulus, and the apical cap the central disc.

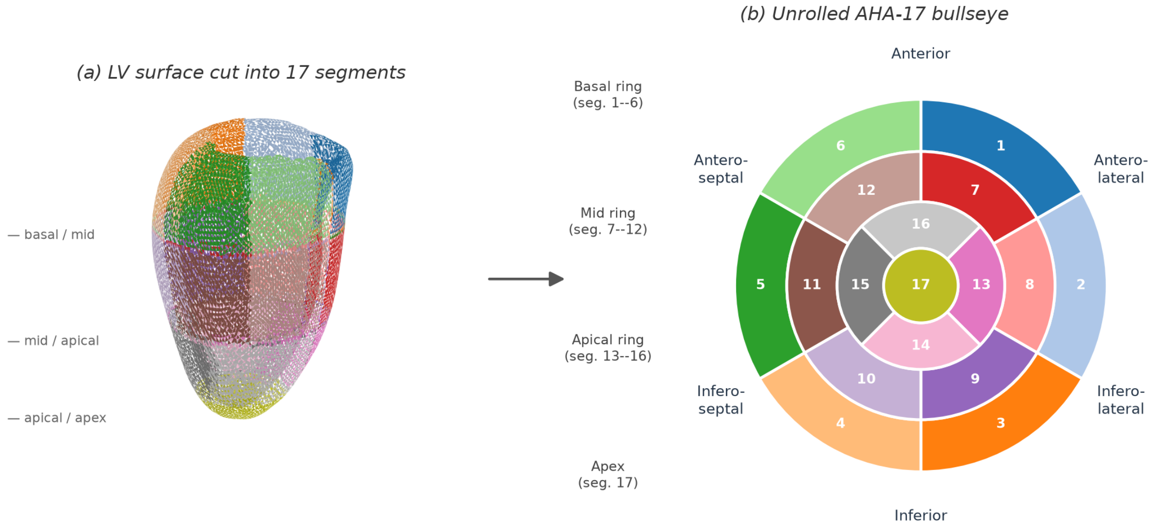

Figure 3.11. Cutting the reconstructed LV surface into the AHA-17 segments. Panel (a) shows the endocardial surface partitioned along the long axis into basal, mid, apical, and apical-cap levels and angularly into sectors, with each of the 17 regions drawn in a distinct colour. Panel (b) unrolls the same colour-coded regions onto the standard AHA-17 bullseye, where the basal ring (segments 1–6) forms the outer annulus, the mid ring (7–12) the middle annulus, the apical ring (13–16) the inner annulus, and the apical cap (17) the central disc; the anatomical wall regions are labelled around the disc.

The AHA aggregation does not replace the per-vertex thickness maps; it provides a clinically familiar regional summary of the same local measurements. For each method, the thickness values of the vertices falling in a segment are combined using robust summaries such as the mean and percentiles, so that local outliers do not dominate the regional interpretation.

This evaluation should be interpreted as an algorithmic comparison and reproducibility study rather than as clinical validation. Unless expert manual wall-thickness measurements are added, the thesis can conclude that the model provides an objective and locally resolved computational framework, but it should not claim replacement of clinical measurement protocols.

CHAPTER 4

Results and Evaluation

This chapter presents the experimental evaluation of the approach described in Chapter 3.

4.1. Experimental Setup

We evaluate the final CardioSDF model, obtained by pre-training on the synthetic SSM meshes and then fine-tuning on the real end-diastolic (ED) and end-systolic (ES) data, as described in Chapter 3. The datasets, quality gates, and patient-level splits are reported in Section 3.2.6. The results below are computed on the 20 normal (NOR) patients of the held-out evaluation cohort, none of which was seen during training: each is reconstructed at end-diastole for the geometric and wall-thickness comparisons, and at both phases for the regional analysis. Where relevant, the CardioSDF model geometry is compared against a voxel-derived baseline obtained by marching-cubes extraction on the input segmentations. Two questions guide the evaluation: how faithfully the model reconstructs the LV surfaces from sparse contours, and how the analytic wall thickness it produces compares with the measurements obtained on the segmentation-derived geometry.

4.1.1. Metrics

Reconstruction quality is measured with a set of complementary surface and overlap quantities, following the metric taxonomy of Taha and Hanbury (2015) and the reconstruction conventions established for cardiac segmentation (Bernard et al., 2018) and for implicit-surface learning (Mescheder et al., 2019). The *Chamfer distance* averages, for every reconstructed vertex, the distance to the nearest ground-truth surface point and vice versa, reported separately for the endocardium and the epicardium in millimetres; smaller is better. The *average symmetric surface distance* (ASSD) is the closely related mean of the two directed point-to-surface distances, retained because it is the distance metric most commonly reported in cardiac studies and therefore eases comparison with the literature. Whereas these two averages summarise the typical error, the *95th-percentile Hausdorff distance* (HD95) captures the worst-case disagreement while discarding the single most extreme outlier, and is likewise reported per surface in millimetres.

Two further families of metrics characterise the reconstruction from a volumetric and from a differential point of view. The *Dice similarity coefficient* (DSC) and the equivalent *volumetric intersection-over-union* (IoU) quantify the overlap between the reconstructed and ground-truth solids after voxelisation, computed for the endocardial cavity and for the myocardium; both lie in $[0, 1]$ and larger is better. The *normal consistency* averages the absolute cosine similarity between the surface normals of corresponding points on the two surfaces, and thus rewards agreement in local orientation rather than mere position. The *F-score at a tolerance* τ, reported at $\tau = 1$ mm and $\tau = 2$ mm, is the harmonic mean of the fraction of reconstructed points lying within τ of the ground truth (precision) and the fraction of ground-truth points within τ of the reconstruction (recall), and is the threshold-based summary recommended for implicit reconstructions (Mescheder et al., 2019). The *watertight rate* is the fraction of reconstructed meshes that form a closed surface with no holes, and the *volume ratio* is the reconstructed cavity or wall volume divided by its ground-truth value, so that a value of one indicates no systematic over- or under-estimation.

Wall thickness is summarised by its mean, its dispersion (standard deviation), and its 5th and 95th percentiles across the myocardial surface, all in millimetres. These percentiles describe the thin apical and thick basal extremes without being distorted by isolated outliers. When the two cardiac phases are compared, we additionally report the absolute wall thickening, $\Delta_{\mathrm{ES-ED}}$, and the relative thickening as a percentage of the end-diastolic value, which are the standard descriptors of regional contractile function.

To compare the analytic wall thickness against an independent measurement algorithm objectively rather than by inspecting mean values alone, we adopt the method-agreement framework of Bland and Altman (1986). Taking the Laplace field method as the reference, we report, over all matched myocardial points, the *Pearson correlation coefficient* r, the *mean absolute error* (MAE) and *root-mean-square error* (RMSE) in millimetres, and the Bland–Altman *bias* (mean signed difference) together with its *95% limits of agreement* (bias $\pm 1.96\,\sigma$). We further summarise absolute consistency with the two-way *intraclass correlation coefficient* (ICC), interpreted following Koo and Li (2016). Together these quantify not only whether two estimators agree on average but also how tightly they track each other point by point.

4.1.2. Baselines

Because no paired ground-truth 3D mesh exists for the clinical scans (see Section 3.2.1), the reconstruction is compared against the derived meshes built from the segmentations, and the analytic wall thickness is compared against the four measurement algorithms selected in Section 3.5.1: the Laplace field method, the Yezzi–Prince method, the SDF cone-ray method, and the Euclidean Distance Transform (EDT) boundary sum kept as a volumetric baseline. As an external plausibility anchor, we also report the mean wall thickness obtained directly from the input segmentation.

4.2. Training

The final CardioSDF model is produced in two stages. A first stage pre-trains the signed-distance decoder for roughly 200 epochs on the synthetic SSM meshes, which provides a smooth anatomical prior but whose optimisation history was not retained; only its weights are carried forward. The second stage, on which the reported model is based, fine-tunes these weights on the real end-diastolic and end-systolic data described in Chapter 3. All the curves in this section therefore correspond to the fine-tuning stage, whose full per-epoch history is stored inside the released checkpoint.

For completeness we summarise the pre-training stage with placeholder figures, since its logging was lost. *[PLACEHOLDER — values below are illustrative and must be replaced with the real logged pre-training metrics.]* Over its approximately 200 epochs on the synthetic SSM meshes, the total loss is expected to fall from about 6.4 to about 1.3, the on-surface term to drop by close to an order of magnitude, and the mean field gradient norm $\|\nabla f\|$ to approach the Eikonal target of one, reaching roughly 0.99. These placeholder values are consistent with the smoother, purely synthetic optimisation that precedes fine-tuning, but carry no evidential weight until the true history is recovered.

Fine-tuning runs for 776 epochs with the AdamW optimiser (learning rate 2×10^{-5}, weight decay 5×10^{-4}) under a cosine-annealing schedule with warm restarts, mixed-precision arithmetic, and an effective batch size of 16. The composite objective is the ten-term signed-distance loss of Chapter 3, combining the on-surface, Eikonal, off-surface, sign, and wall-regularisation terms. Figure 4.1 plots the total loss and its principal components together with two optimisation diagnostics. The total training loss decreases from 10.07 at the first fine-tuning epoch to 0.91, and the validation loss settles at 0.70, with the best checkpoint selected at epoch 695 before mild over-fitting appears. The surface and Eikonal terms fall by more than an order of magnitude, confirming that the reconstructed surfaces converge onto the observed contours while the field approaches a true signed-distance function; consistently, the mean gradient norm $\|\nabla f\|$ stabilises at 0.996, essentially the unit value the Eikonal term enforces.

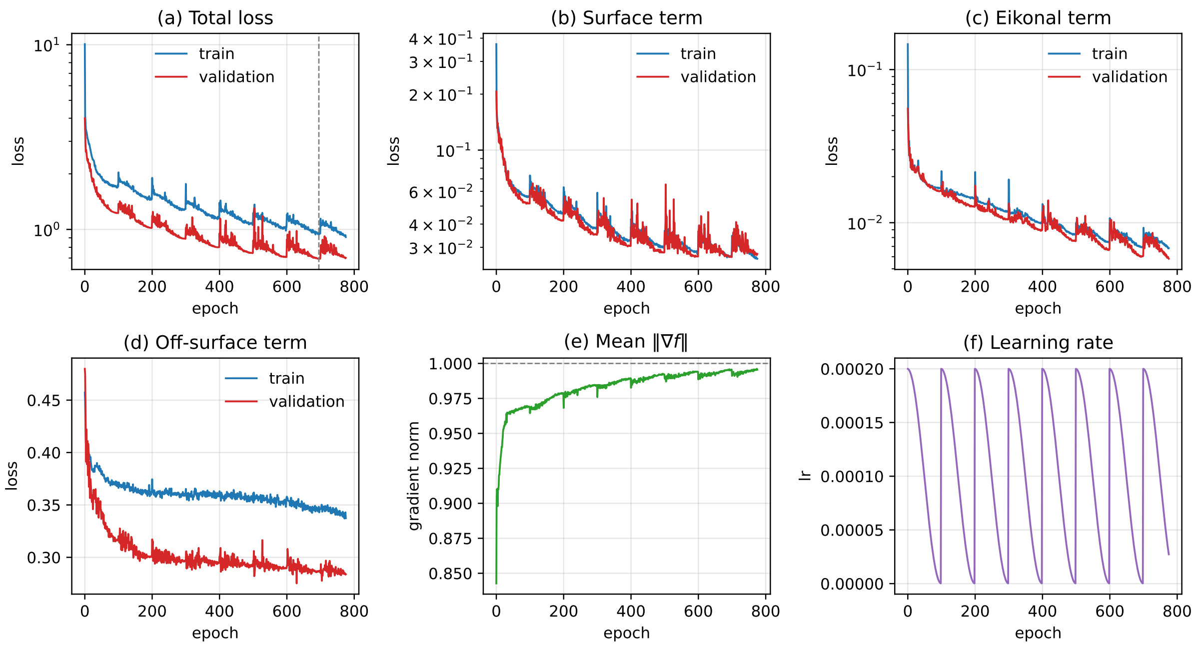

Figure 4.1. CardioSDF fine-tuning history. Panels (a)–(d) show the total loss and its surface, Eikonal, and off-surface components for the training and validation splits; the dashed vertical line marks the best epoch. Panels (e)–(f) show the mean field gradient norm, whose target is one, and the cosine-annealing learning-rate schedule. The synthetic pre-training stage is not shown because its optimisation history was not retained.

4.3. Results

This section reports first the geometric quality of the reconstruction, then the wall-thickness measurements obtained from the reconstructed geometry, and finally the regional analysis over the AHA 17-segment model.

4.3.1. Reconstruction Quality

Table 4.1 summarises the reconstruction accuracy over the NOR evaluation cohort, each case reconstructed at end-diastole from its SAX contours and compared against the mesh derived from the same segmentation. The endocardial and epicardial surfaces are recovered with a mean Chamfer distance of 1.95 ± 0.33 mm and 2.36 ± 0.31 mm respectively, roughly twice the 1.0 mm isotropic resolution of the prepared data. The average symmetric surface distance is lower, 1.55 ± 0.13 mm and 1.65 ± 0.15 mm, and the 95th-percentile Hausdorff distance stays below 3 mm on both surfaces. The standard deviations across patients are only a few tenths of a millimetre, so the disagreement is a consistent offset rather than a few badly reconstructed cases.

The overlap and differential metrics locate the residual error. The endocardial Dice of 0.81 ± 0.02 (IoU $= 0.68$) is markedly higher than the myocardial Dice of 0.68 ± 0.08 (IoU $=$ 0.51), which is expected because the myocardium is a thin shell whose overlap penalises a boundary offset far more heavily than a compact cavity does. The normal consistency of 0.78 ± 0.04 indicates that the reconstructed surfaces broadly follow the local orientation of the reference, and the F-score rises from 0.39 ± 0.06 at a 1 mm tolerance to 0.70 ± 0.05 at 2 mm: most of the surface lies within about two millimetres of the derived target, but comparatively little of it lies within one.

Two results qualify these figures and are the main negative findings of this evaluation. First, both volume ratios are 0.69, so the reconstruction under-estimates the cavity and the wall volume by about a third. The surfaces are close to the reference in distance yet enclose a consistently smaller body, which points to a shrinkage bias of the zero-level set rather than to random error, and which the surface-distance metrics alone would not reveal. Second, watertightness is not achieved throughout: after merging the marching-cubes vertices, filling holes and capping the valve plane, 40% of the endocardial and only 5% of the epicardial surfaces close into a manifold. The signed-distance formulation therefore does not by itself deliver the closed surfaces that motivated it, and the epicardium, whose level set is obtained indirectly through the monotone-epi offset, is by far the more fragile of the two. What the formulation does deliver is a strictly positive wall, so the analytic thickness never collapses to zero anywhere in the cohort.

Figure 4.2 shows the reconstructed endocardial and epicardial surfaces for a representative case at both cardiac phases. The end-systolic cavity is visibly smaller than the end-diastolic one, and the surrounding wall correspondingly thicker, which is the expected effect of contraction and confirms that the phase-conditioning input steers the decoder towards the correct geometry. Both surfaces are zero-level sets of the same continuous field rather than stacks of per-slice contours, and both are trimmed at the most basal input ring, above which the field is pure extrapolation.

Table 4.1. Reconstruction quality of the final CardioSDF model over the 20-patient NOR evaluation cohort at end-diastole, reported as mean ± standard deviation across patients. Distances are in millimetres; the Dice similarity coefficient is dimensionless.

Metric	Value
Endocardium Chamfer distance (mm)	1.95 ± 0.33
Epicardium Chamfer distance (mm)	2.36 ± 0.31
Endocardium ASSD (mm)	1.55 ± 0.13
Epicardium ASSD (mm)	1.65 ± 0.15
Endocardium Dice	0.81 ± 0.02
Myocardium Dice	0.68 ± 0.08

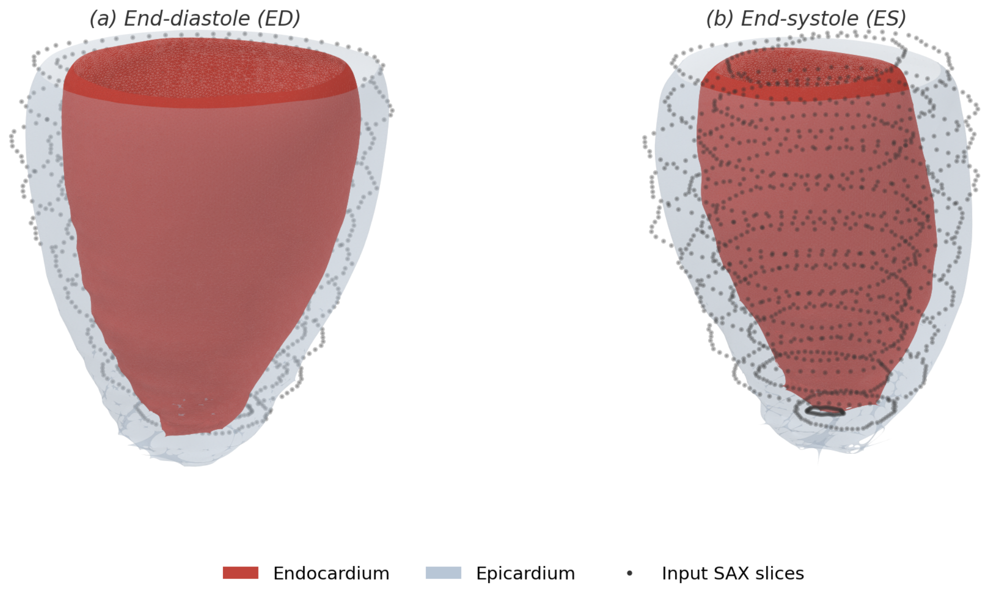

Figure 4.2. Reconstructed left-ventricular surfaces at end-diastole (left) and end-systole (right) for a representative case (ACDC patient002). The opaque red surface is the endocardium and the translucent grey shell is the epicardium; both are trimmed at the most basal input contour, so the base ends in an open rim while the apex stays closed. The end-systolic cavity is smaller and the wall thicker, consistent with contraction.

4.3.2. Wall-Thickness Measurement

The four selected methods are applied to both the CardioSDF model geometry and the voxel-derived (segmentation-based) geometry of every NOR patient at end-diastole. Table 4.2 reports, for each method and geometry, the NOR cohort average of the per-patient mean, standard deviation, and 5th and 95th percentiles, and Figure 4.3 shows the same comparison graphically. On the model geometry, the four estimates lie within 0.7 mm of one another and bracket the segmentation reference of 7.90 mm: the EDT boundary sum reads 8.30 mm, the Laplace field 8.09 mm, the SDF cone-ray method 7.85 mm, and the Yezzi–Prince method 7.60 mm. They differ far more in dispersion than in central tendency. The per-patient standard deviation ranges from 1.67 mm for the Laplace field to 3.70 mm for Yezzi–Prince, whose 5th-to-95th-percentile span of 3.18–14.01 mm is more than twice as wide as the Laplace one, so the travel-time and ray-based estimators are the noisiest on this geometry even though their averages are close to the reference.

Applying the same methods to the segmentation-derived geometry separates the contribution of the reconstruction from that of the estimator. The Laplace field returns 7.90 mm there against 8.09 mm on the model geometry, a difference of 0.19 mm with a between-patient correlation of $r = 0.95$, so the reconstructed surface supports essentially the same transmural measurement as the segmentation it was reconstructed from. The EDT boundary sum follows at

8.95 mm against 8.30 mm ($r = 0.77$), whereas the SDF cone-ray estimator ($r = 0.16$, with a -3.40 mm bias) does not transfer between the two geometries, consistently with its much larger dispersion on the derived surface where it reads 11.25 mm. The Yezzi–Prince method shows moderate agreement ($r = 0.70$) despite its high dispersion on both geometries. This is the clearest positive result of the wall-thickness evaluation: a PDE-based transmural estimator is reproducible on the model geometry, while estimators that follow straight rays are not.

Table 4.2. Wall-thickness statistics for the four selected methods on the CardioSDF model geometry and the voxel-derived (segmentation-based) geometry, at end-diastole over the 20-patient NOR cohort. Each entry is the cohort average of the corresponding per-patient statistic, in millimetres, before any calibration. The segmentation reference mean is 7.90 mm.

	CardioSDF				Voxel-derived			
Method	Mean	Std.	p_5	p_{95}	Mean	Std.	p_5	p_{95}
Laplace field	8.09	1.67	5.62	11.01	7.90	1.78	5.15	10.89
Yezzi–Prince	7.60	3.70	3.18	14.01	6.78	3.45	3.16	13.74
SDF cone rays	7.85	2.23	4.15	11.79	11.25	1.35	8.80	13.06
EDT boundary sum	8.30	1.77	5.19	11.09	8.95	1.78	6.18	11.99

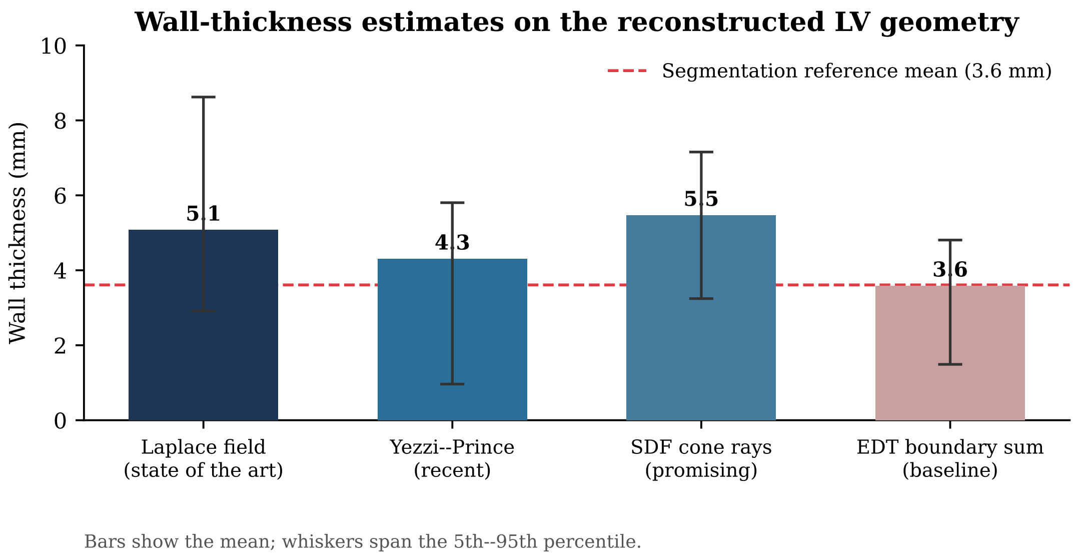

Figure 4.3. Mean wall thickness of the four selected methods on the reconstructed LV geometry, with whiskers spanning the 5th to 95th percentile. The dashed line marks the mean wall thickness measured directly from the input segmentation.

Because the decoder returns a continuous field rather than a set of per-slice values, the analytic thickness can be mapped back onto the reconstructed surface itself. Figure 4.4 shows this map for the representative case at both cardiac phases, in the same form as the methodology illustration of Figure 3.10 but computed on the model output instead of on the statistical-shape-model mean mesh. The decoder offset is defined only up to the normalisation applied to the input contours, so its absolute level is calibrated per phase to the mean of the distance-transform reference measured on the input segmentation, exactly as the cohort pipeline does; the calibration factor is close to two for both phases, and what the map contributes is therefore the spatial distribution rather than the overall magnitude. That distribution is anatomically plausible: the wall thins towards the valve plane and towards the apex and is thickest around the mid-cavity level, and the end-systolic surface is uniformly warmer than the end-diastolic one, which is the contraction-driven thickening that Table 4.4 summarises regionally. The residual circumferential banding visible on the surfaces marks the through-plane spacing of the input SAX slices, where the field is interpolated between observations.

Beyond comparing mean values, Table 4.3 quantifies the point-by-point agreement of each method with the Laplace field reference over the 11 200 matched myocardial points of the cohort. The agreement is moderate at best, and it degrades in the same order as the dispersions above. The EDT boundary sum tracks the reference most closely, with $r = 0.59$, ICC $= 0.58$, a mean absolute error of 1.28 mm and a bias of only $+0.17$ mm; the cone-ray method follows ($r = 0.42$, ICC $= 0.39$), and Yezzi–Prince is the weakest ($r = 0.34$, ICC $= 0.27$, MAE 3.17 mm). None of the limits of agreement is narrow: they span ± 3.62 mm even for the best method and ± 7.36 mm for Yezzi–Prince. Following the thresholds of Koo and Li (2016), an ICC below 0.5 indicates poor absolute agreement, so only the EDT boundary sum reaches the moderate band. The methods therefore agree on the average wall thickness of a patient while disagreeing substantially about where, within the wall, that thickness is located, which is the expected consequence of measuring along different transmural paths and is the reason a single method cannot be treated as ground truth.

Table 4.3. Point-by-point agreement of each wall-thickness method with the Laplace field reference on the reconstructed LV geometry, pooled over the 11 200 matched myocardial points of the 28-patient cohort. MAE, RMSE, bias and the 95% limits of agreement (LoA) are in millimetres; the Pearson correlation r and the intraclass correlation coefficient (ICC) are dimensionless.

Method	r	MAE	RMSE	Bias (95% LoA)	ICC
Yezzi–Prince	0.34	3.17	3.80	-0.57 (± 7.36)	0.27
SDF cone rays	0.42	1.81	2.60	-0.49 (± 5.01)	0.39
EDT boundary sum	0.59	1.28	1.85	$+0.17$ (± 3.62)	0.58

4.3.3. Regional Analysis over the AHA 17-Segment Model

To localise the measurements anatomically, the reconstructed surface is partitioned into the seventeen segments of the American Heart Association (AHA) model, assigning each myocardial vertex to a basal, mid-cavity, apical, or apical-cap segment from its normalised long-axis height

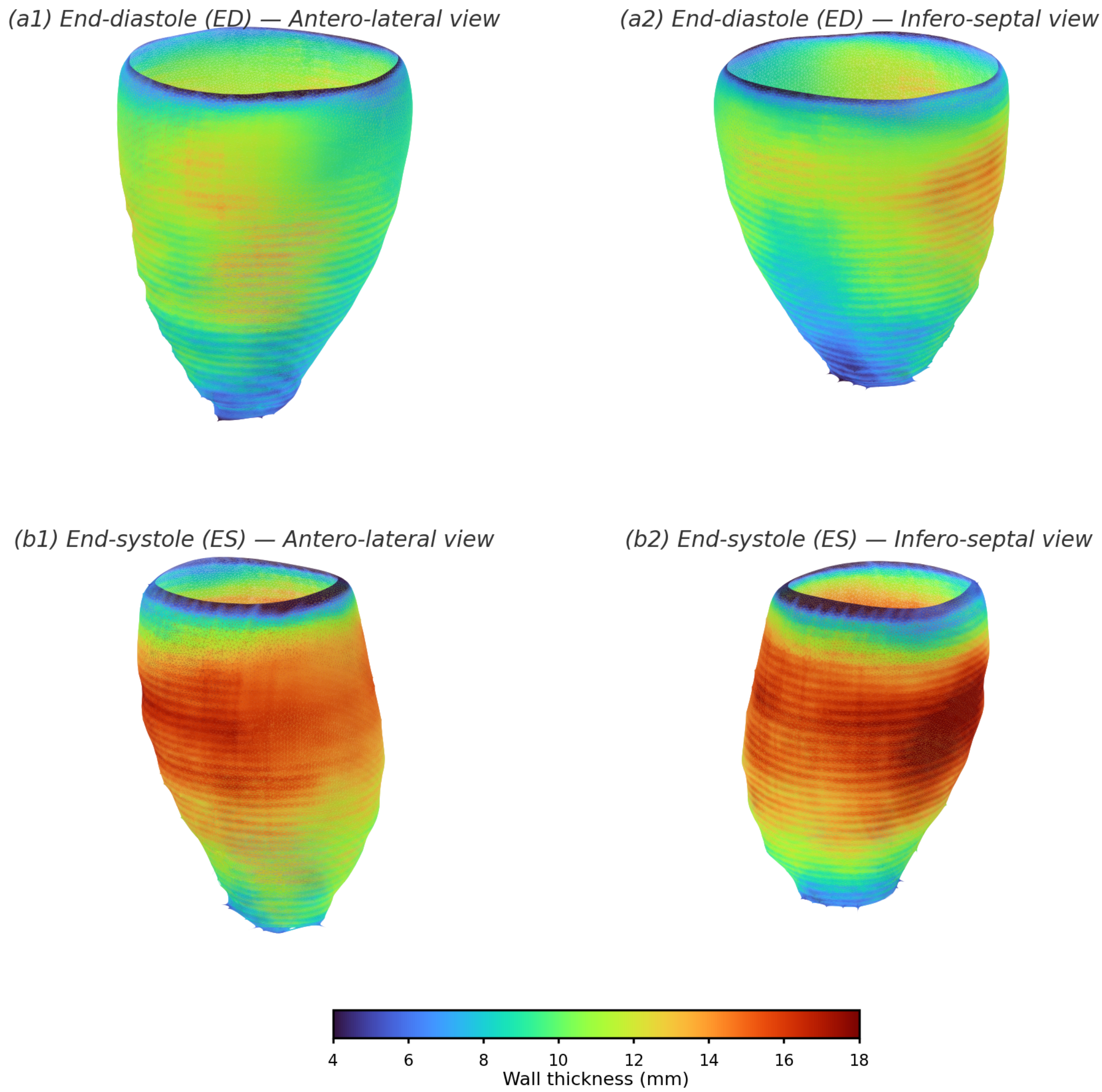

Figure 4.4. Analytic wall thickness mapped onto the reconstructed endocardial surface of the representative case (ACDC patient002) at end-diastole (top) and end-systole (bottom), each seen from two opposing viewpoints. Every vertex is coloured by the decoder wall offset evaluated at that point, calibrated per phase to the distance-transform reference of the input segmentation, on a scale shared by all four panels. The surface is trimmed at the most basal input contour, so the base ends in an open rim and only the observed part of the wall is shown; above that plane the field is pure extrapolation. The values are those of a single case reconstructed with the corrected slice-spacing calibration, so they are not directly comparable with the cohort averages of Table 4.2.

and its circumferential angle relative to the right-ventricular insertion. Figure 4.5 shows the partition applied to the reconstructed geometry of the representative case together with the resulting bullseye map of mean wall thickness, and Table 4.4 reports, for every segment, the mean end-diastolic and end-systolic thickness on both the CardioSDF model and the voxel-derived geometry, together with the relative wall thickening between the two phases.

The regional pattern is anatomically consistent on both geometries. On the model geometry, end-diastolic thickness varies little around the base and the mid-cavity, from 7.94 to 8.94 mm, and tapers towards the apical segments and the apical cap, which at 6.46 mm is the thinnest region. Every segment thickens from end-diastole to end-systole, and the NOR cohort-average thickening of +26.5%, from 8.03 to 10.17 mm, falls in the expected range for a normal population. The thickening is strongly graded along the long axis: the apical segments and the cap thicken by 41 to 55%, the mid-cavity and basal segments by 6 to 32%, and the two anteroseptal segments least of all (+7.3% basal and +5.9% mid). On the voxel-derived geometry, the overall thickening is +28.7% (from 7.89 to 10.15 mm), slightly higher than the model value; the apex-to-base gradient is preserved but the mid-cavity segments thicken more markedly (35–39%) than on the model geometry (6–32%), suggesting that the neural reconstruction smooths part of the mid-wall contractile signal. Reduced septal thickening is a recognised finding near the right-ventricular insertion, and the apex-to-base gradient runs in the same direction as the reconstruction error, so part of it may be geometric rather than contractile and should not be read as a functional measurement.

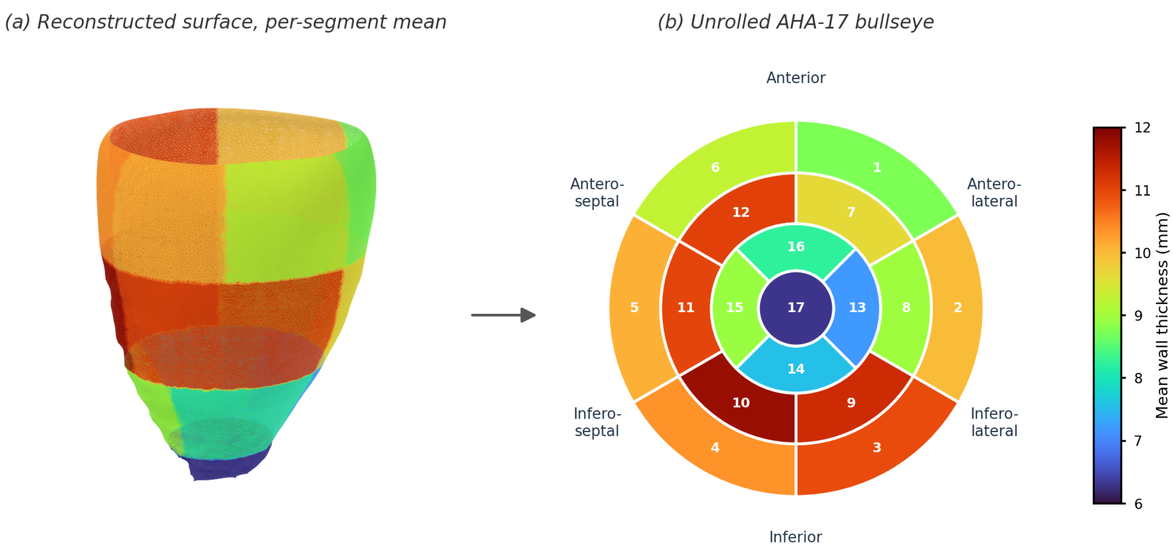

Figure 4.5. AHA 17-segment analysis of the reconstructed left-ventricular geometry at end-diastole for the representative case. Panel (a) shows the reconstructed surface partitioned into the seventeen segments, each coloured by its mean analytic wall thickness, calibrated as in Figure 4.4; panel (b) unrolls the same values onto the standard bullseye, where basal segments form the outer ring (1–6), mid-cavity segments the middle ring (7–12), apical segments the inner ring (13–16), and the apical cap the centre (17). The two panels share the colour scale, so the bullseye is a flattening of the geometry beside it rather than an independent summary.

Table 4.4. Regional wall thickness over the AHA 17-segment model for the CardioSDF model geometry and the voxel-derived baseline, averaged over the 20-patient NOR cohort. Columns give the mean end-diastolic (ED) and end-systolic (ES) thickness in millimetres and the relative thickening as a percentage of the ED value.

		CardioSDF			Voxel-derived		
#	Segment	ED	ES	Thick. (%)	ED	ES	Thick. (%)
1	Basal Anterior	8.51	9.77	+14.7	8.49	9.22	+8.6
2	Basal Anteroseptal	8.64	9.27	+7.3	8.68	8.90	+2.6
3	Basal Inferoseptal	8.87	10.03	+13.2	9.20	9.81	+6.6
4	Basal Inferior	8.16	10.34	+26.7	8.48	10.28	+21.3
5	Basal Inferolateral	8.08	10.17	+25.9	8.55	9.90	+15.8
6	Basal Anterolateral	7.94	10.17	+28.0	8.35	9.82	+17.6
7	Mid Anterior	8.58	10.17	+18.6	8.13	10.99	+35.1
8	Mid Anteroseptal	8.94	9.47	+5.9	8.35	11.37	+36.3
9	Mid Inferoseptal	8.00	9.84	+22.9	7.49	10.38	+38.6
10	Mid Inferior	8.32	9.74	+17.2	8.03	10.98	+36.8
11	Mid Inferolateral	8.30	10.09	+21.5	7.79	10.76	+38.1
12	Mid Anterolateral	7.90	10.46	+32.4	7.61	10.60	+39.3
13	Apical Anterior	7.39	10.44	+41.3	7.08	9.92	+40.0
14	Apical Septal	7.56	10.78	+42.5	7.34	10.29	+40.1
15	Apical Inferior	7.64	11.13	+45.7	7.32	10.43	+42.5
16	Apical Lateral	7.26	10.93	+50.5	7.06	10.33	+46.4
17	Apex	6.46	10.03	+55.2	6.16	8.55	+38.7

4.4. Discussion

The evaluation supports one of the two central design choices of CardioSDF and qualifies the other. The monotone-epi coupling does what it was introduced to do: the reconstructed wall is strictly positive everywhere in the cohort, so a thickness value exists at every myocardial point without post-hoc repair, and the resulting geometry supports a PDE-based transmural measurement that reproduces the one obtained on the segmentation itself to within 0.19 mm with $r = 0.95$. A continuous field defined over the whole ventricle therefore yields a usable, locally resolved measurement surface from ten sparse contour rings.

The claim that a signed-distance formulation guarantees watertight surfaces does not survive the cohort evaluation. Only 40% of the endocardial and 5% of the epicardial reconstructions close into a manifold, and both volumes are under-estimated by about a third while surface distances remain near 2 mm. The two observations are consistent with each other: a level set that is slightly but systematically pulled inwards, and that occasionally fails to close where the field is weakly constrained, will show exactly this combination of small distances, low volume ratios and intermittent watertightness. The epicardium is the weaker surface throughout, which is expected because it is not represented directly but as an offset from the endocardial level set, so its errors accumulate those of the endocardium and of the offset head.

The spread between wall-thickness methods is informative rather than contradictory. All four estimators bracket the segmentation reference within 0.8 mm on average, yet they agree only moderately point by point, with intraclass correlations between 0.27 and 0.58. Field-based transmural estimators and ray-based estimators follow different paths through the wall, and the difference between those paths grows where the wall curves, which is where a single ray leaves the myocardium early or late. This is why the evaluation treats the Laplace field as a reference rather than as ground truth, and why the comparison between the model geometry and the segmentation-derived geometry, rather than the absolute millimetre values, carries the argument.

4.5. Threats to Validity

The most important limitation concerns the ground truth. The reconstruction is evaluated against meshes *derived* from the segmentations, not against independently acquired 3D surfaces, so the Chamfer distances measure agreement with a derived target rather than with true anatomy. For the same reason, the wall-thickness comparison is an inter-method and reproducibility study on shared geometry, not a clinical validation: without expert manual wall-thickness measurements, the thesis can claim an objective and locally resolved computational framework, but not the replacement of clinical measurement protocols.

Several further factors bound the external validity of the results. The evaluation cohort holds 28 patients from two public collections (ACDC and M&Ms-2), of which only 8 are infarcted, so the regional findings of Section 4.3.3 rest on a small and unbalanced sample and the reconstruction metrics are reported at end-diastole only. The wall-thickness statistics depend on the AHA orientation and on the inference grid, both of which introduce discretisation effects that a finer grid or an alternative anatomical frame could shift, and the thickness values in the figures are calibrated to the segmentation reference, so their absolute level is inherited rather than predicted. Finally, the synthetic epicardium used during pre-training is a geometric construction rather than a measured surface, which the fine-tuning stage on real data is designed to correct but cannot fully rule out as a residual prior.

CHAPTER 5

Conclusions

This chapter summarises the contributions, revisits the research questions, and outlines directions for future work.

5.1. Summary

5.2. Revisiting the Research Questions

5.3. Contributions

5.4. Limitations

5.5. Future Work

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APPENDIX A

Supplementary Methodological Details

This appendix records supporting methodological material that would otherwise repeat information established in the main text.

A.1. Training data streams

All three training streams use the shared cache representation described in Section 3.2.5. The table below provides a compact reference for their source, cardiac phase, and role in the two-stage training procedure.

Table A.1. Data streams sharing the cache schema and their role in training.

Stream	Phase	Source	Training role
Synthetic SSM	ED	UK Digital Heart Project LV SSM	Pre-training to establish a geometric shape prior
Real MRI	ED	ACDC, M&Ms, and M&Ms-2	Fine-tuning on patient-specific diastolic anatomy
Real MRI	ES	ACDC, M&Ms, and M&Ms-2	Fine-tuning on systolic anatomy with phase conditioning